# INS 3121
# SOIL MECHANICS

## Compressibility of Soils

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# Introduction

- The compression is caused by
    a. Deformation of soil particles
    b. Relocations of soil particles
    c. Expulsion of water or air from the void spaces

# Introduction

- Soil Settlements
  1. Elastic settlement (or Immediate settlement)

     elastic deformation of dry soil and of moist and saturated soils without any change in the moisture content. Immediate settlement equations derived from the theory of elasticity.
  2. Primary consolidation settlement

     a volume change in saturated cohesive soils because of expulsion of the water that occupies the void spaces.
  3. Secondary consolidation settlement

     The result of the plastic adjustment of soil fabrics

# Introduction

- **Total settlement**
  1. Elastic or immediate settlement ($S_e$)
  2. Consolidation settlement ($S_c$)
  3. Secondary settlement ($S_s$)

$$\boldsymbol{S_T = S_e + S_c + S_s}$$

# Elastic Settlement

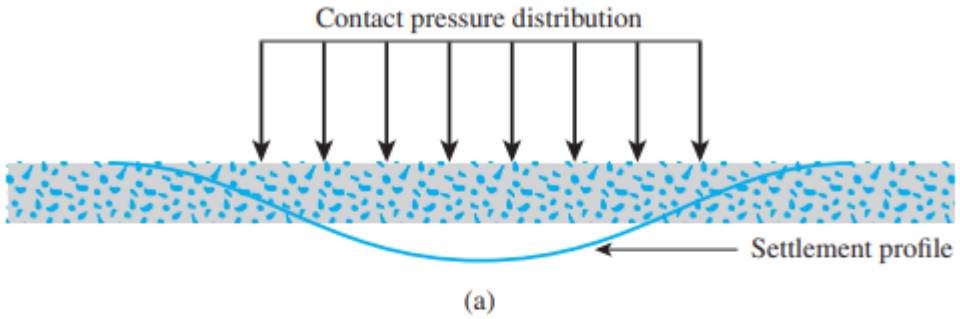



***Figure 11.1***
Elastic settlement profile and contact pressure in clay: (a) flexible foundation; (b) rigid foundation

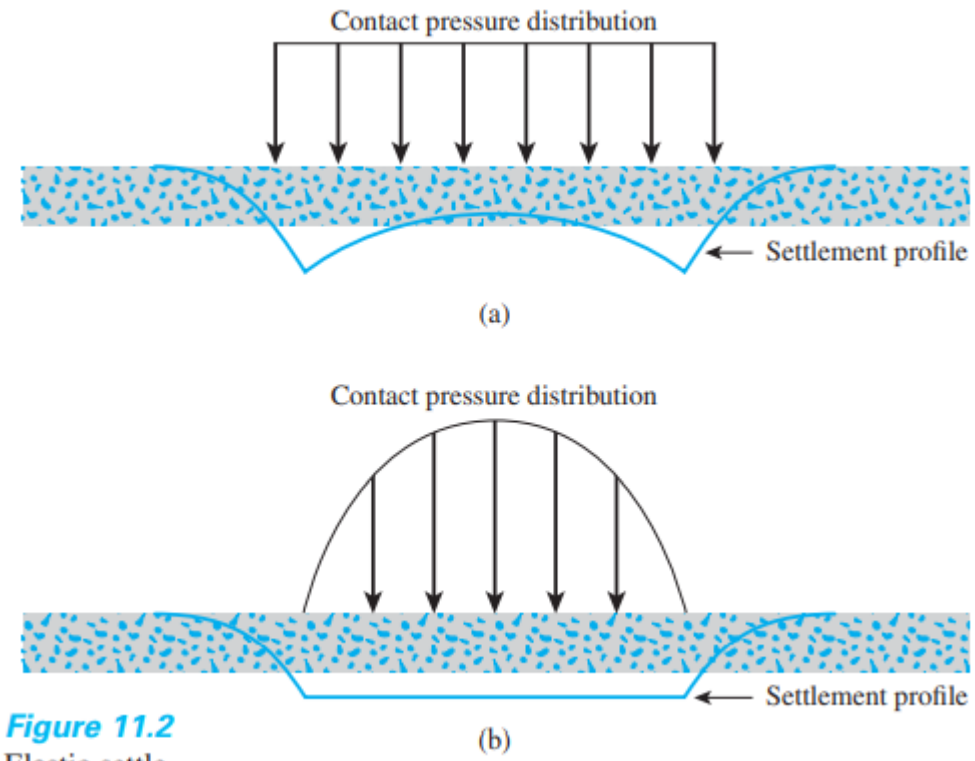



***Figure 11.2***
Elastic settlement profile and contact pressure in sand: (a) flexible foundation; (b) rigid foundation

# CONSOLIDATION SETTLEMENT

- When a saturated soil layer is subjected to a stress increase, the pore water pressure is suddenly increased.
- Because of rapid drainage of the pore water in sandy soils, immediate settlement and consolidation take place simultaneously.
- When a saturated compressible clay layer is subjected to a stress increases, elastic settlement occurs immediately.
- The excess pore water pressure generated by loading gradually dissipates over a long period of time.
- The settlement caused by consolidation in clay may be several times greater than the immediate settlement.

# CONSOLIDATION SETTLEMENT

## t = 0 and valve is closed

- A simple model of the time-dependent deformation of saturated clayey
- Let the inside area of the cross section of the cylinder be equal to A. The cylinder is filled with water and has a frictionless watertight piston and valve
- The excess hydrostatic pressure :$\Delta u = \frac{P}{A}$

$$P = P_s + P_w$$

$P_s$= load carried by the spring and $P_w$ = load carried by the water.

Valve closed → $P_s = 0$ and $P_w = P$

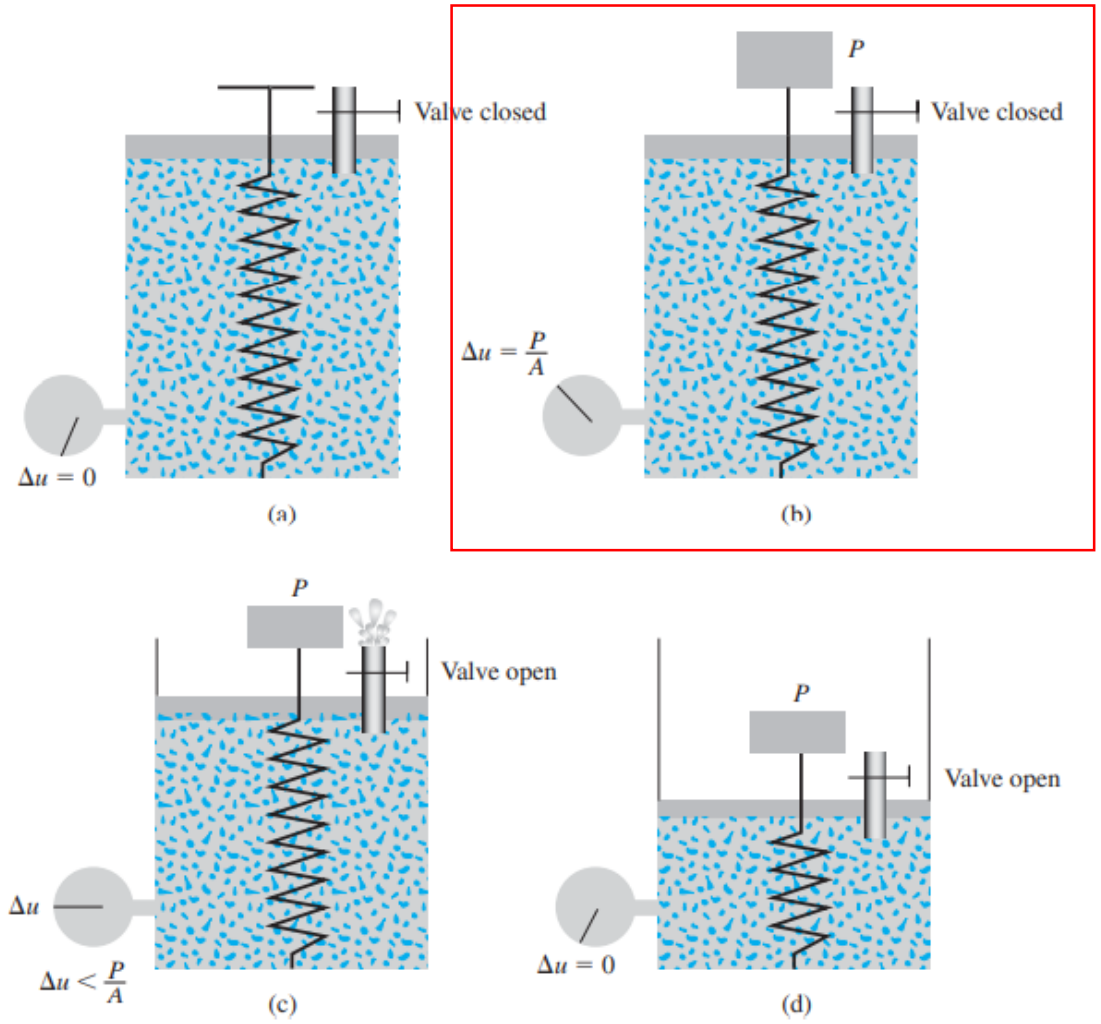



*Figure 11.5* Spring-cylinder model

# CONSOLIDATION SETTLEMENT

## 0<t< ∞ AND valve opened

- If the valve is opened, the water will flow outward.
- This flow will be accompanied by a reduction of the excess hydrostatic pressure and an increase in the compression of the spring.

$$P_s > 0 \; and \; P_w < P \; (that \; is, \Delta u < \frac{P}{A})$$

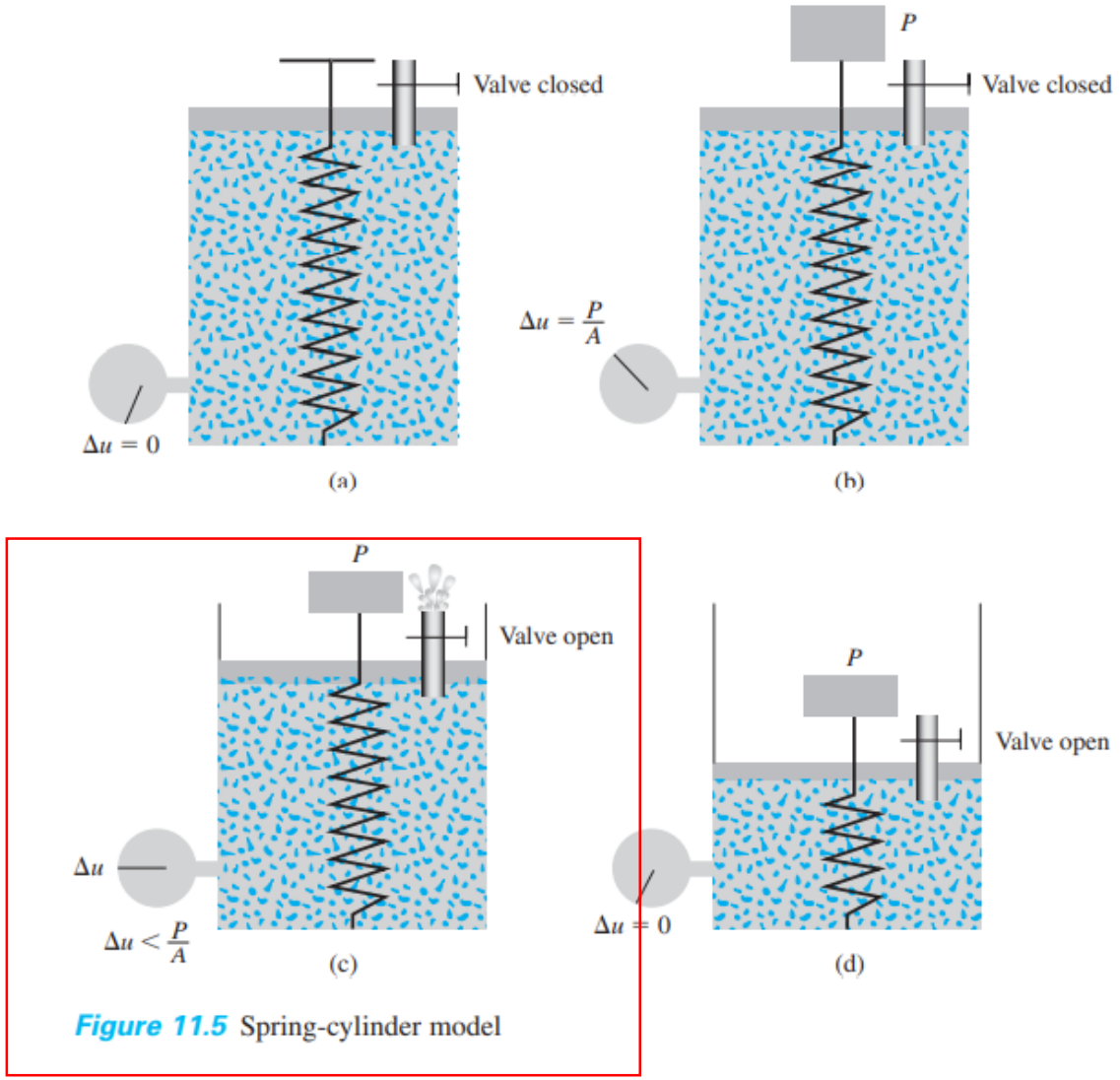



*Figure 11.5* Spring-cylinder model

# CONSOLIDATION SETTLEMENT

## t = ∞ AND valve opened

- After some time, the excess hydrostatic pressure will become zero and the system will reach a state of equilibrium

$$P_s = P \; and \; P_w = P$$

and

$$P = P_s + P_w 0$$

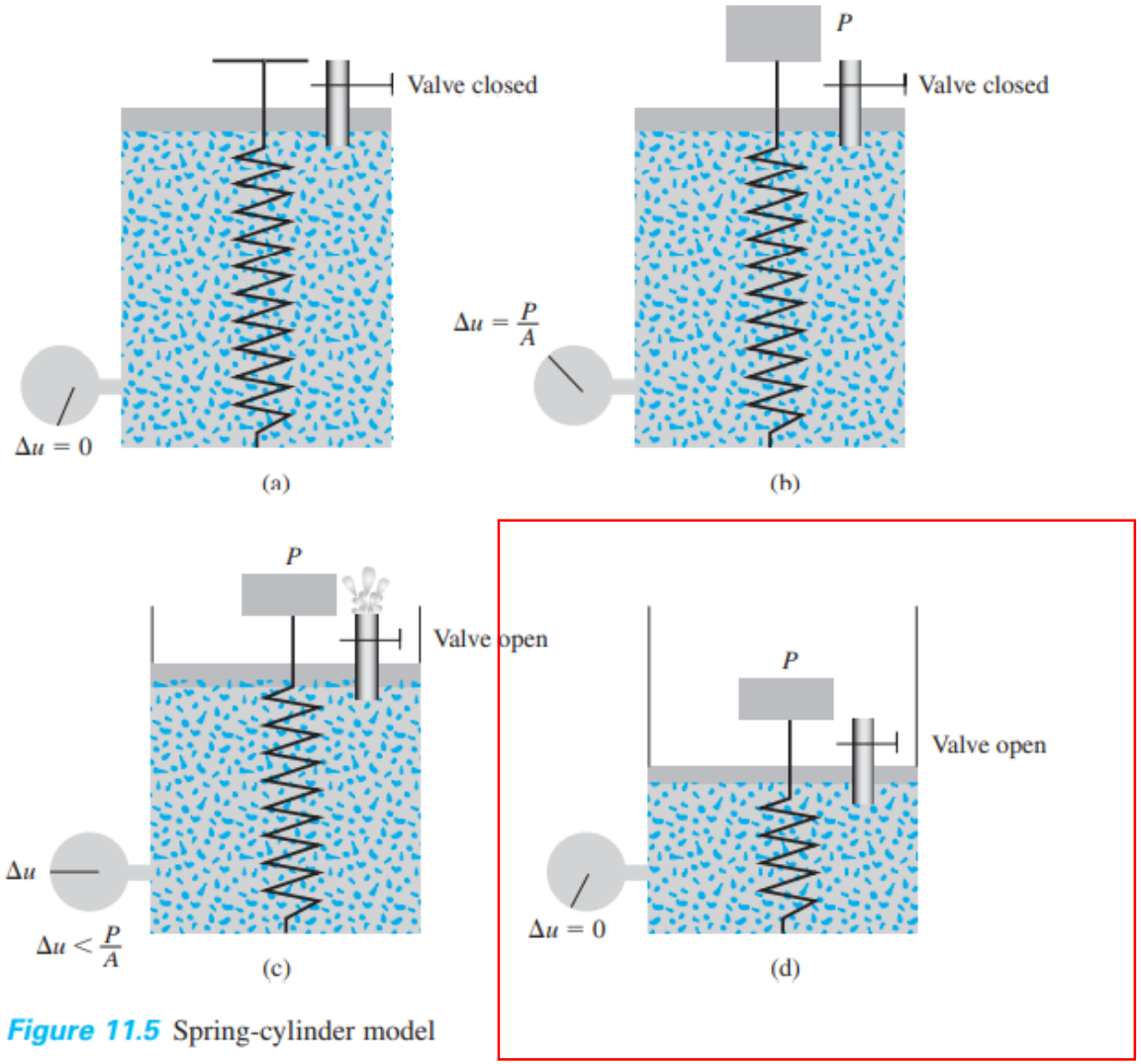
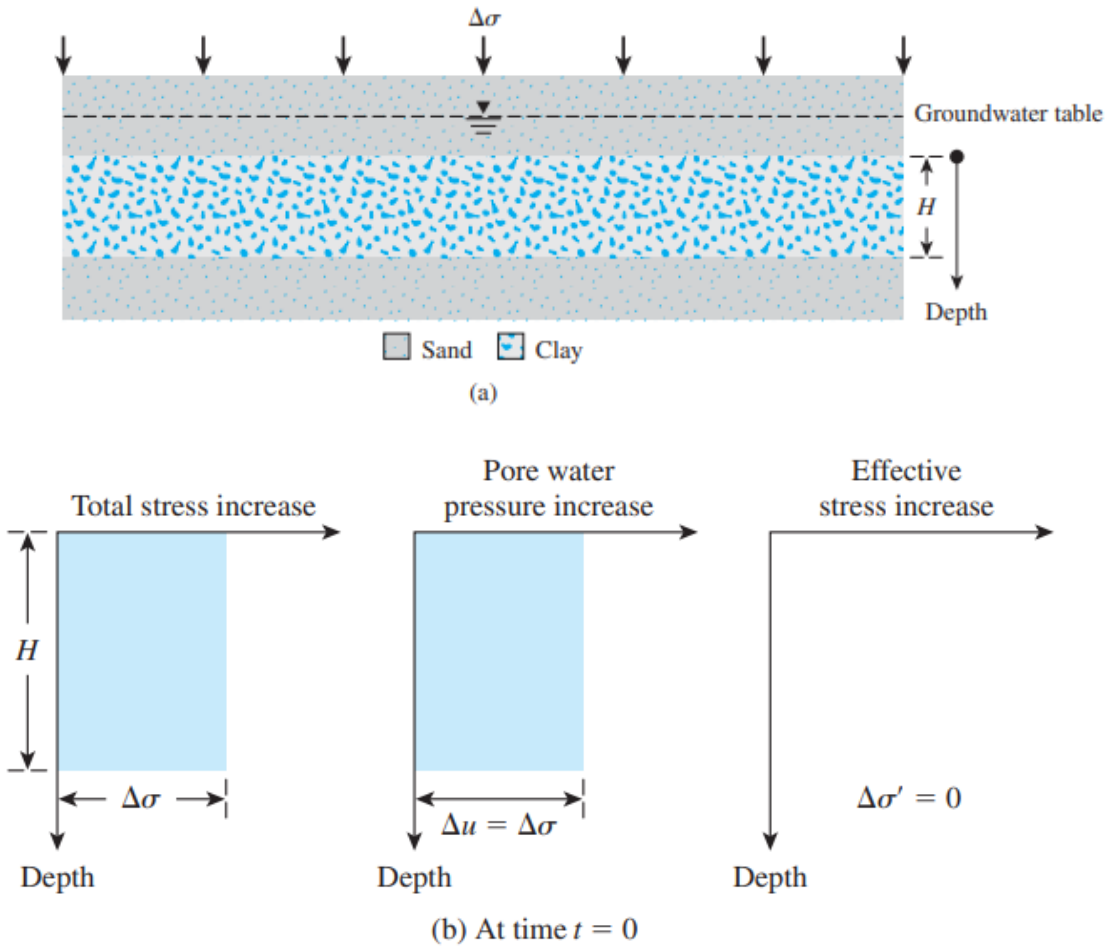



*Figure 11.5* Spring-cylinder model

# CONSOLIDATION SETTLEMENT

**t = 0**

- Since clay has a very low coefficient of permeability and water is incompressible as compared to the soil skeleton, at time t = 0, the entire incremental stress, Δσ, will be carried by water ( Δσ = Δu ) at all depths.



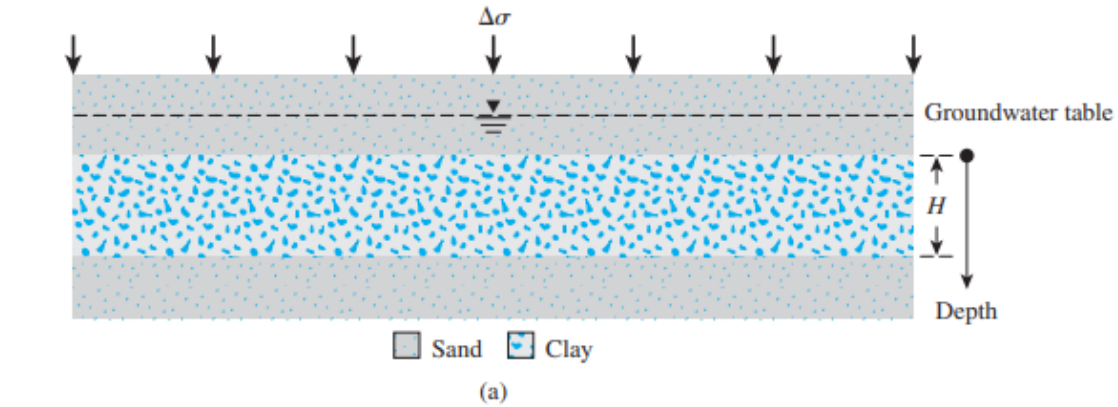
# CONSOLIDATION SETTLEMENT

**0<t < ∞**

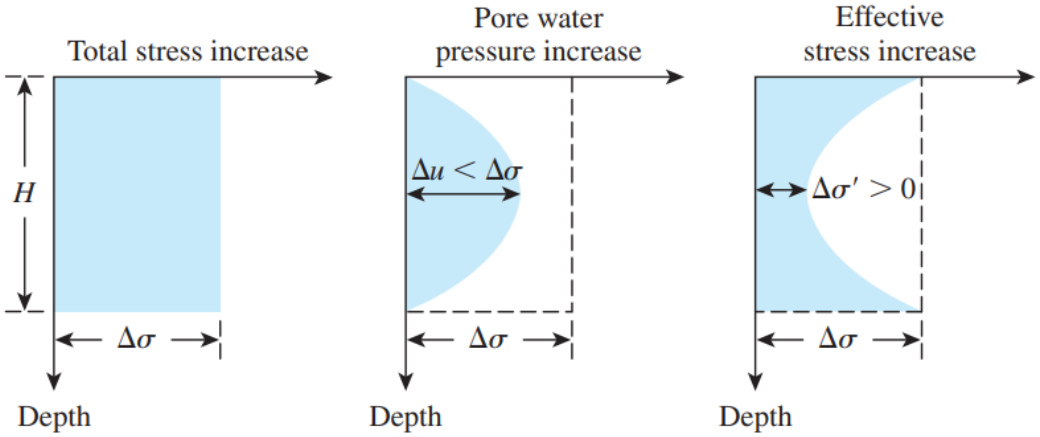
- After the application of incremental stress, Δu, to the clay layer, the water in the void spaces will start to be squeezed out and will drain in both directions into the sand layers.

$$\Delta\sigma = \Delta\sigma' + \Delta u \quad (\Delta\sigma' > 0 \;\; and \;\; \Delta u < \Delta\sigma)$$

- However, the magnitudes of Δσ' and Δu at various depths will change, depending on the minimum distance of the drainage path to either the top or bottom sand layer.



(a)



(c) At time $0 < t < \infty$

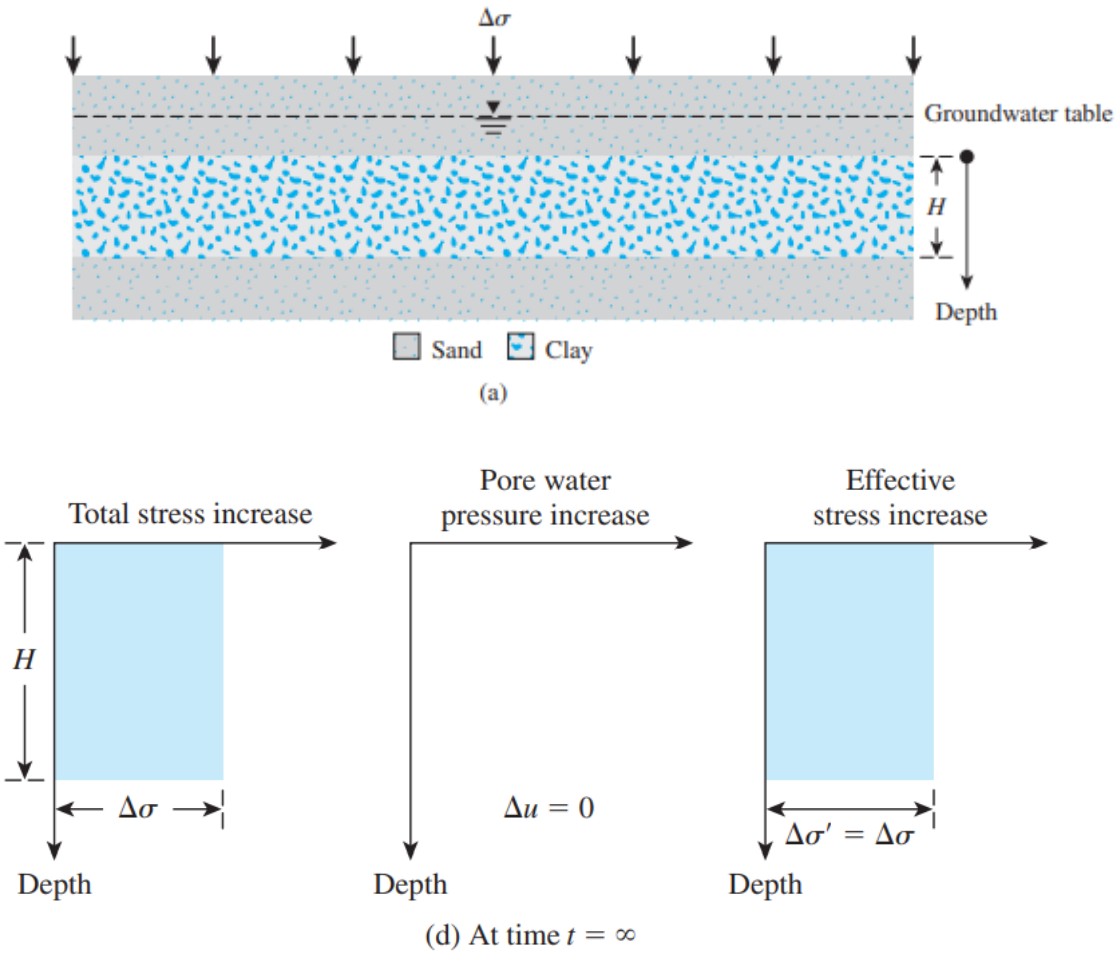
# CONSOLIDATION SETTLEMENT

**t = ∞, theoretically**

- the entire excess pore water pressure would be dissipated by drainage from all points of the clay layer, thus giving Δu = 0.

$$\Delta\sigma = \Delta\sigma'$$

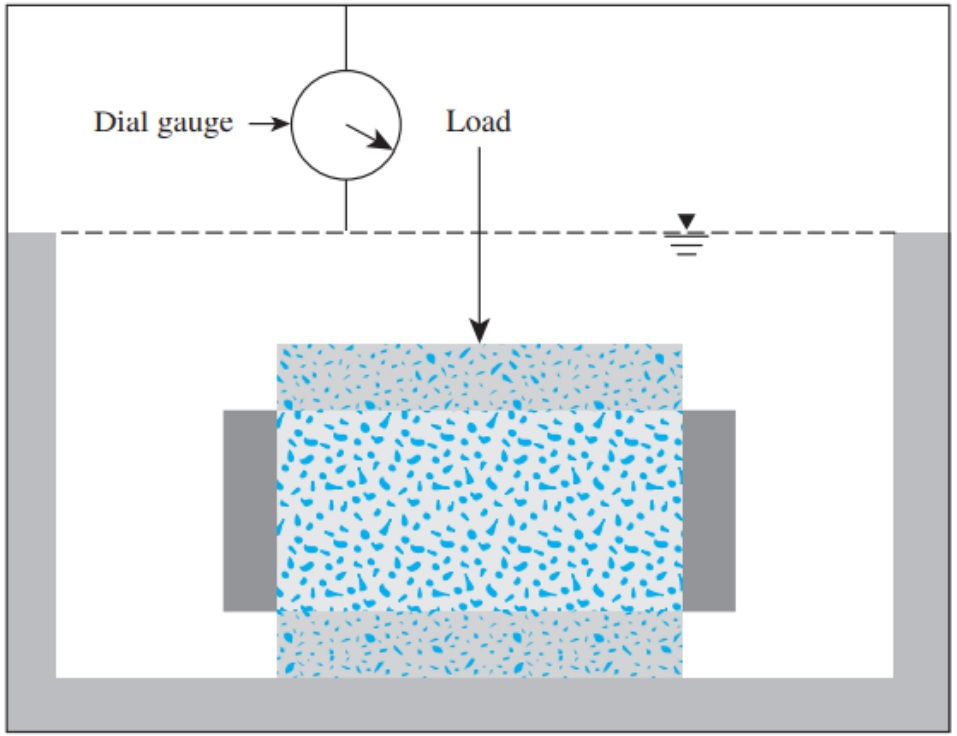
Transfer of excess pore water pressure to effective stress cause the time-dependent settlement in the clay soil layer.



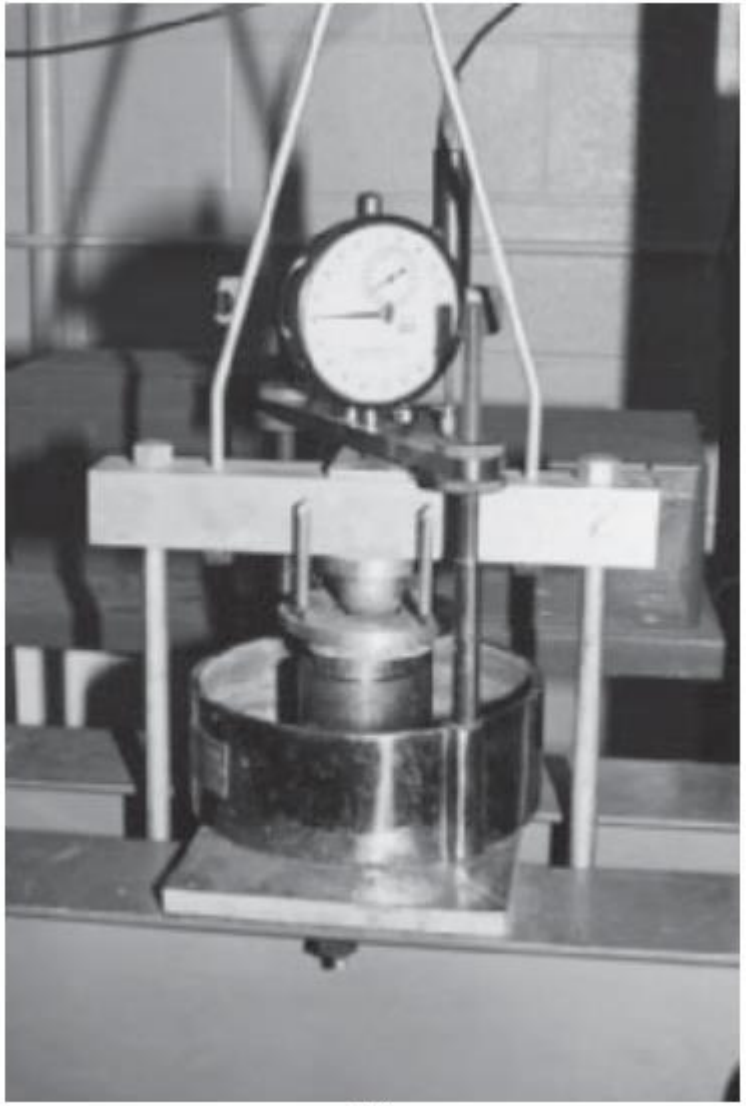
# One-Dimensional Laboratory Consolidation Test

- The one-dimensional consolidation testing procedure was first suggested by Terzaghi
- consolidometer (sometimes referred to as an oedometer)
- The specimens are usually 63.5 mm in diameter and 25.4 mm thick
- Each load is usually kept for 24 hours.
- After that, the load is usually doubled.
- At the end of the test, the dry weight of the test specimen is determined.

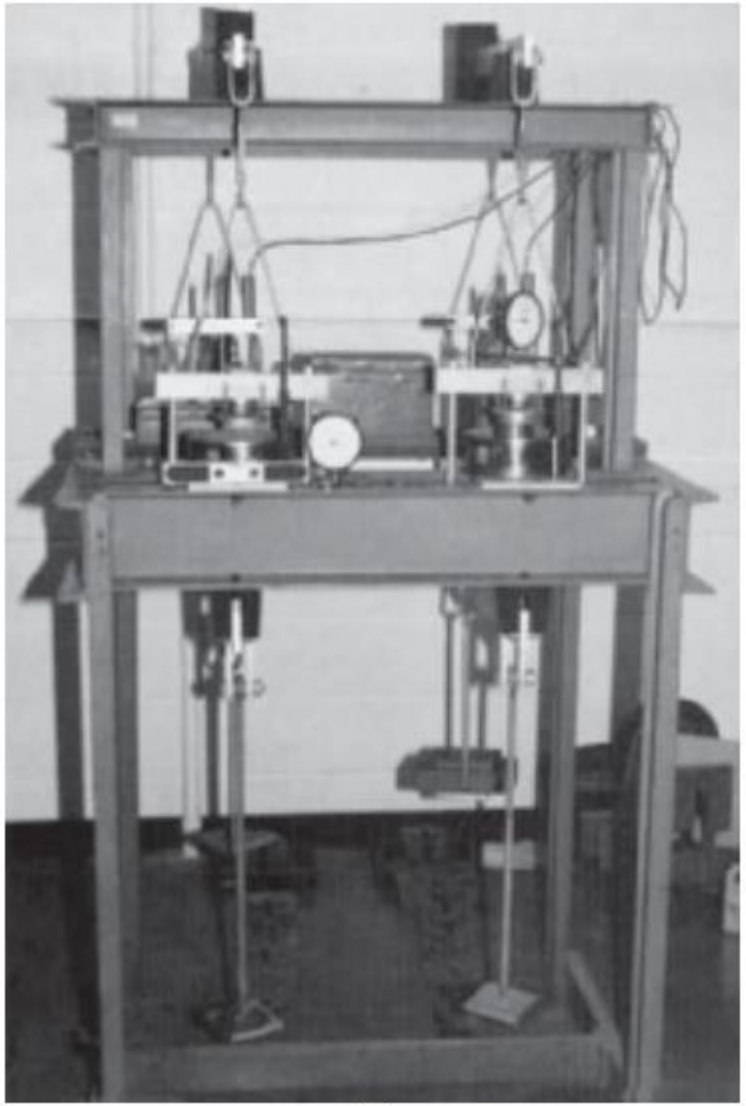
# One-Dimensional Laboratory Consolidation Test




(a)



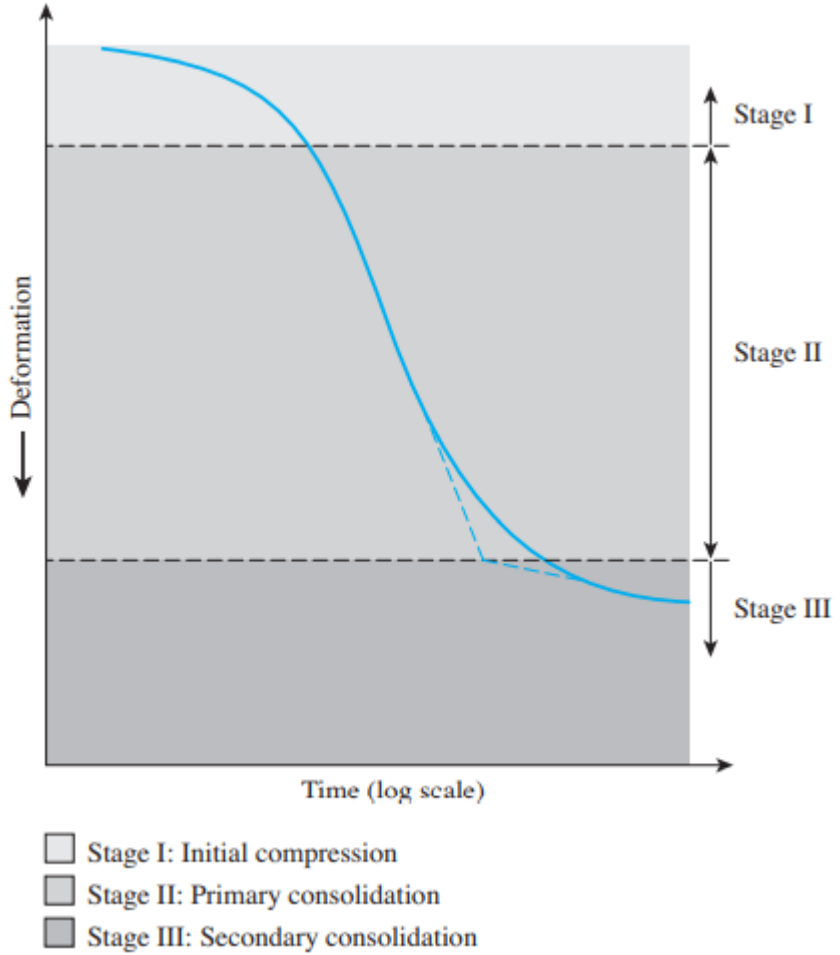
(b)



(c)

# One-Dimensional Laboratory Consolidation Test

- Three distinct stages
  - Stage 1: Initial compression,
    which is mostly caused by preloading.
  - Stage 2: Primary consolidation,
    during which excess pore water pressure gradually is transferred into effective stress because of the expulsion of pore water.
  - Stage 3: Secondary consolidation,
    which occurs after complete dissipation of the excess pore water pressure, when some deformation of the specimen takes place because of the plastic readjustment of soil fabric.



# Void Ratio-Pressure Plots

1. Calculate the height of solids, $H_s$, in the soil specimen

$$H_s = \frac{W_s}{AG_s\gamma_w} = \frac{M_s}{AG_s\rho_w}$$

where $W_s = dry\ weight\ of\ the\ specimen$

$M_s = dry\ mass\ of\ the\ specimen$

$A = area\ of\ the\ specimen$

$G_s = specific\ gravity\ of\ soil\ solids$

$\gamma_w = unit\ weight\ of\ water$

$\rho_w = density\ of\ water$

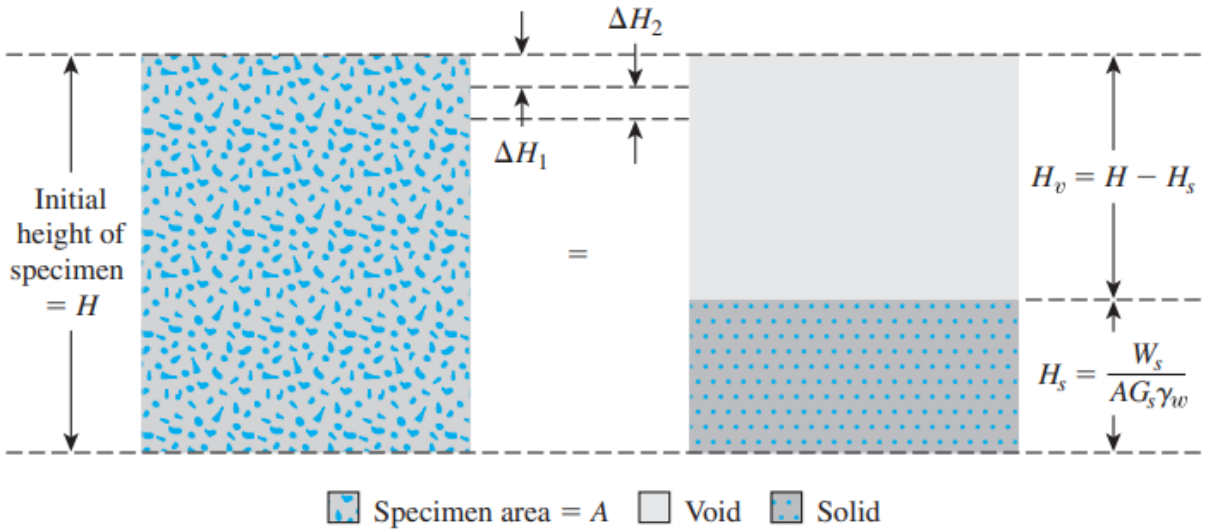



***Figure 11.9*** Change of height of specimen in one-dimensional consolidation test

# Void Ratio-Pressure Plots

2. Calculate the initial height of voids as

$$H_v = H - H_s$$

where H = initial height of the specimen

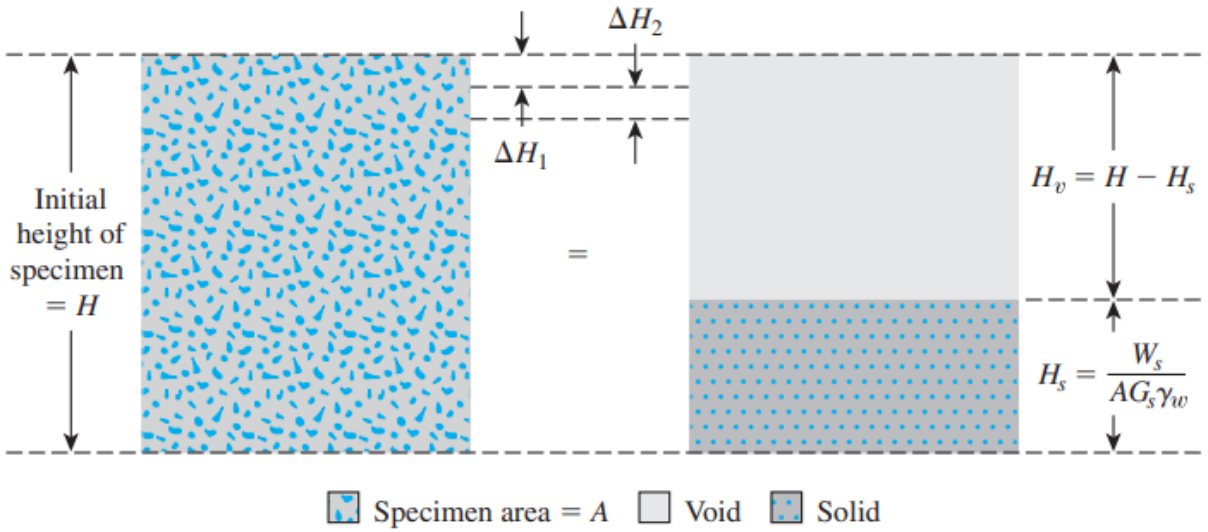



***Figure 11.9*** Change of height of specimen in one-dimensional consolidation test

# Void Ratio-Pressure Plots

3. Calculate the initial void ratio, $e_0$, of the specimen, using the equation

$$e_O = \frac{V_v}{V_s} = \frac{H_v}{H_s}\frac{A}{A} = \frac{H_v}{H_s}$$

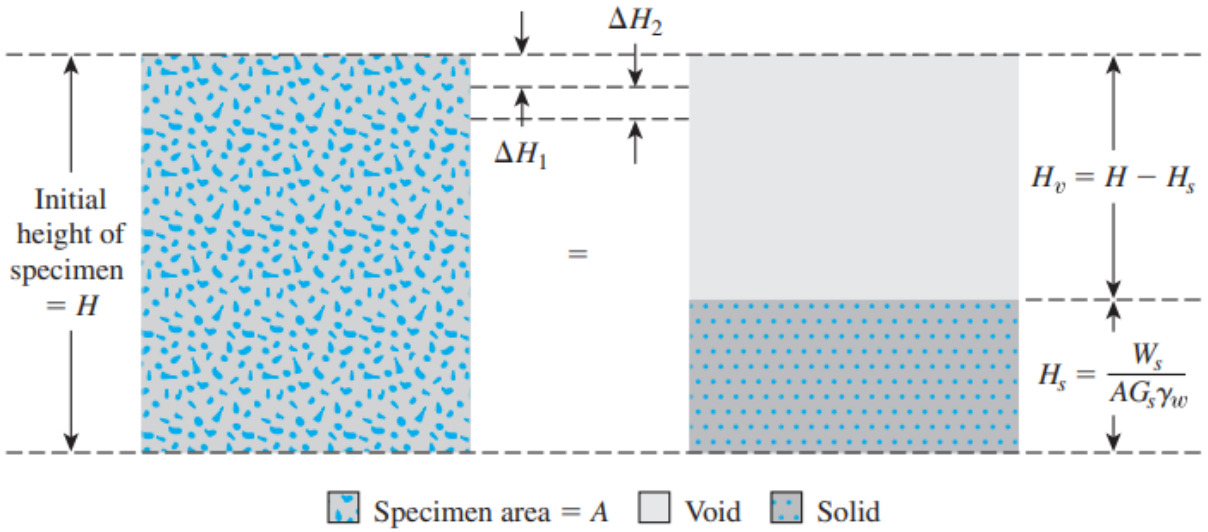



***Figure 11.9*** Change of height of specimen in one-dimensional consolidation test

# Void Ratio-Pressure Plots

4. For the first incremental loading, $\sigma_1$ (total load/unit area of sample), which causes a deformation $\Delta H_1$, calculate the change in the void ratio as

$$\Delta e_1 = \frac{\Delta H_1}{H_s}$$

$\Delta H_1$ is obtained from the initial and the final dial readings for the loading.

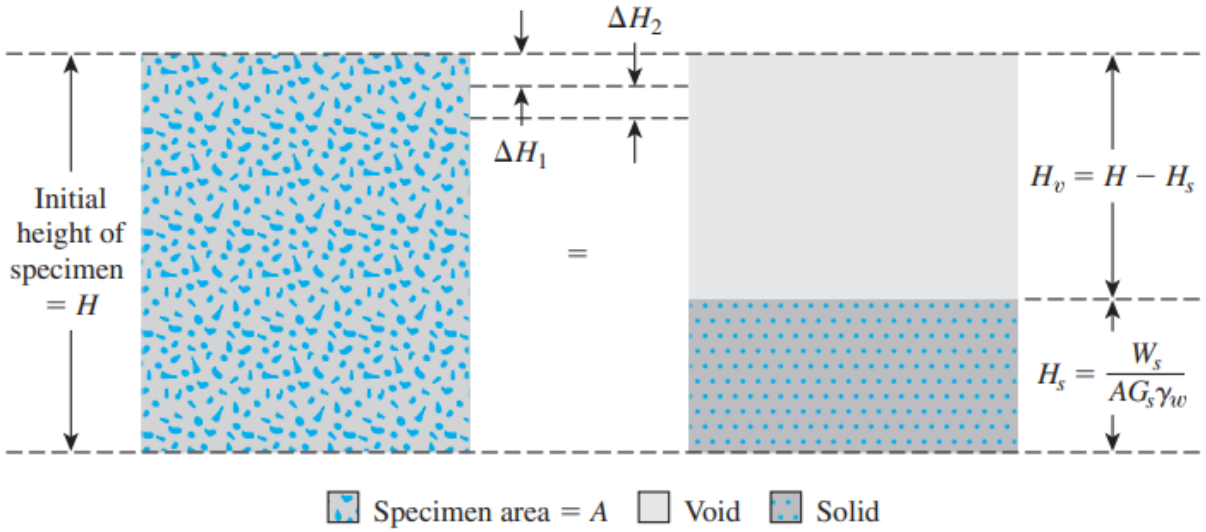



*Figure 11.9* Change of height of specimen in one-dimensional consolidation test

# Void Ratio-Pressure Plots

5. Calculate the new void ratio, $ee$ 1, after consolidation caused by the pressure increment. $e_1 = e_0 - \Delta e_1$

- For the next loading, $\sigma_2$, which causes additional deformation $\Delta H_2$. The void ratio $e_2$ at the end of consolidation can be calculated as

$$e_2 = e_1 - \frac{\Delta H_2}{H_s}$$

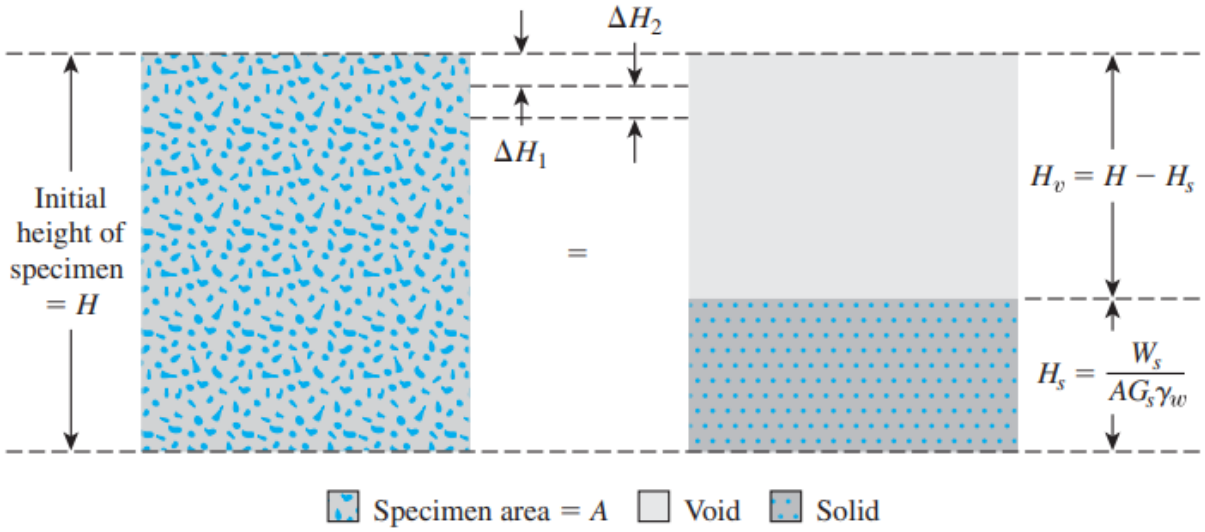



***Figure 11.9*** Change of height of specimen in one-dimensional consolidation test

# Void Ratio-Pressure Plots

- The effective stress $\sigma'$ and the corresponding void ratios (e) at the end of consolidation are plotted on semilogarithmic graph paper

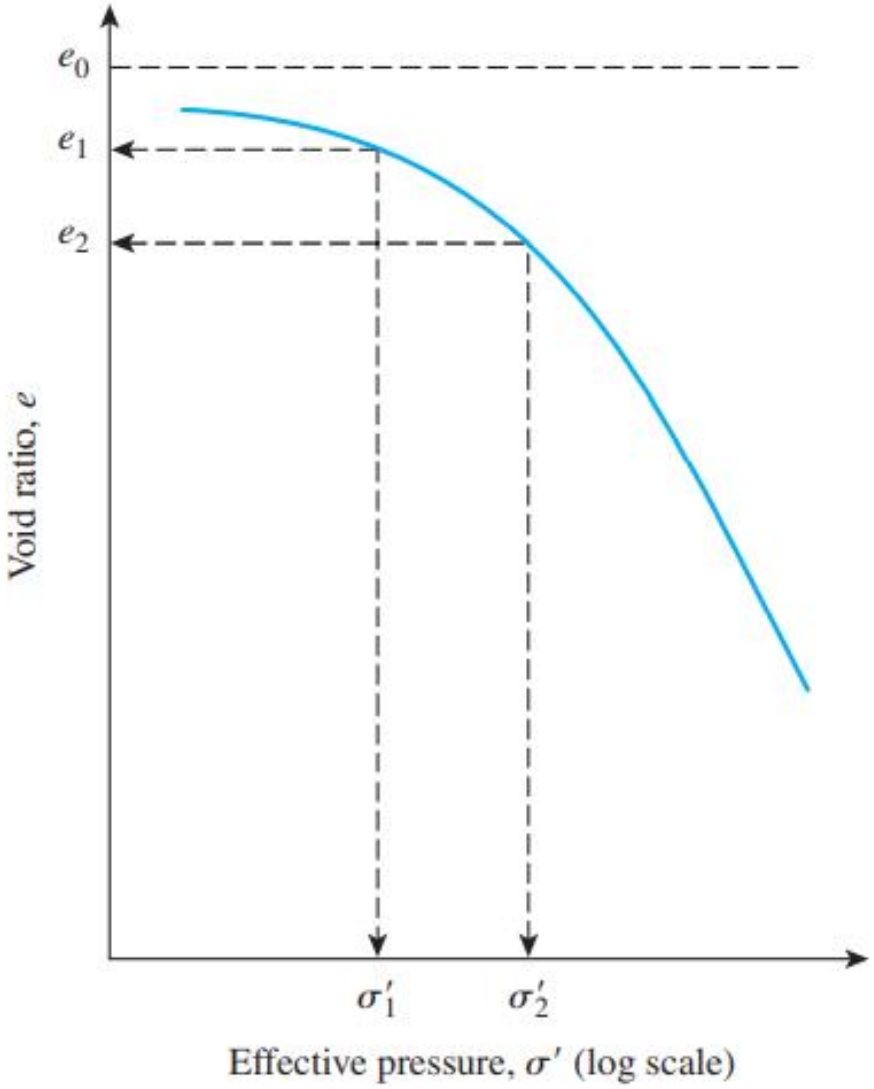



*Figure 11.10* Typical plot of $e$ against log $\sigma'$

# Void Ratio-Pressure Plots

- The upper part of the e-logσ’ plot is somewhat curved with a flat slope, followed by a linear relationship for the void ratio, with logσ’ having a steeper slope.

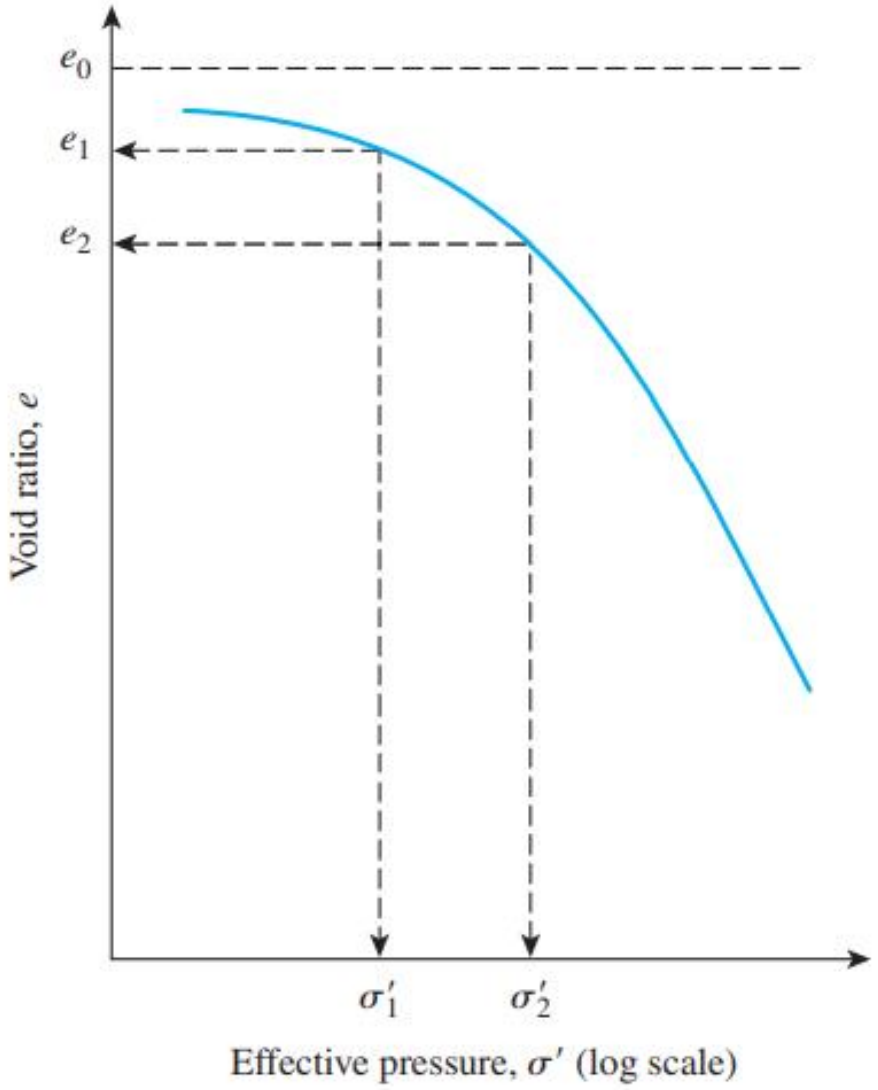



**Figure 11.10** Typical plot of $e$ against log $\sigma'$

# Normally Consolidated and Overconsolidated Clays

- Maximum effective past pressure in its geologic history.

This maximum effective past pressure may be equal to or greater than the existing overburden pressure at the time of sampling.

- When this specimen is subjected to a consolidation test, a small amount of compression (that is, a small change in void ratio) will occur when the total pressure applied is less than the maximum effective overburden pressure in the field to which the soil has been subjected in the past.
- When the total applied pressure on the specimen is greater than the maximum effective past pressure, the change in the void ratio is much larger, and the e – log σ’ relationship is practically linear with a steeper slope.

# Normally Consolidated and Overconsolidated Clays

**1. Normally consolidated**

present effective overburden pressure is the maximum pressure that the soil has been subjected to in the past.

**2. Overconsolidated**

present effective overburden pressure is less than that which the soil has experienced in the past. The maximum effective past pressure is called the preconsolidation pressure

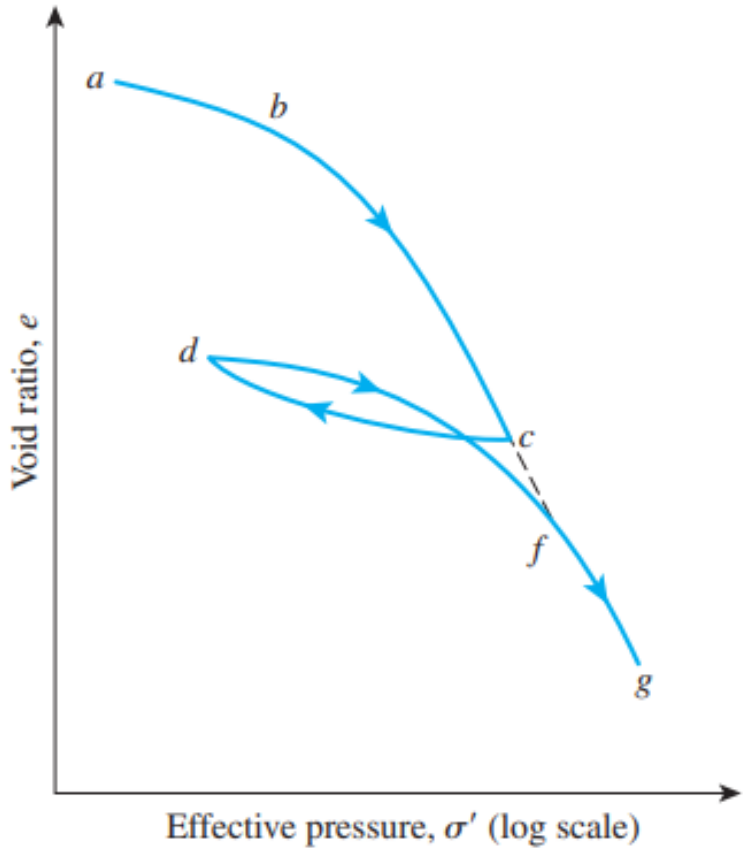



*Figure 11.12* Plot of $e$ against log $\sigma'$ showing loading, unloading, and reloading branches

# Normally Consolidated and Overconsolidated Clays

- **Casagrande (1936)** method to determine the preconsolidation pressure,

1. By visual observation, establish point a at which the e – log σ’ plot has a minimum radius of curvature.
2. Draw a horizontal line ab.
3. Draw the line ac tangent at a.
4. Draw the line ad, which is the bisector of the angle back.

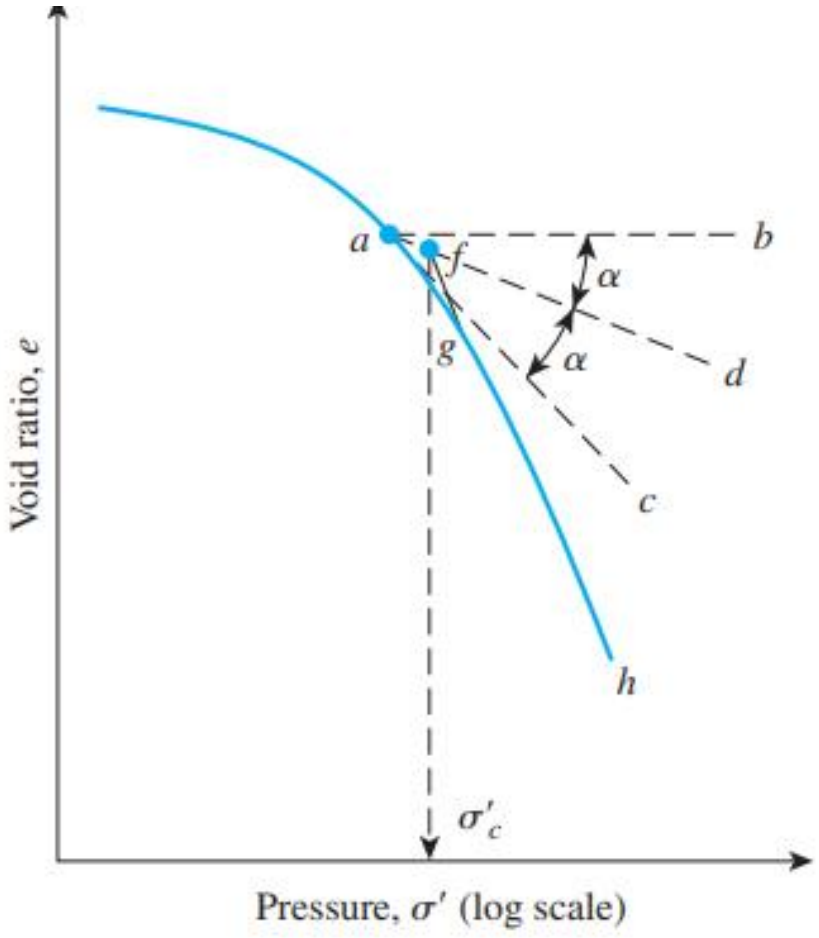



*Figure 11.13* Graphic procedure for determining preconsolidation pressure

# Normally Consolidated and Overconsolidated Clays

- **Casagrande (1936)** method to determine the preconsolidation pressure,

5. Project the straight-line portion gh of the e – log σ’ plot back to intersect ad at f. The abscissa of point f is the preconsolidation pressure, $\sigma'_c$

- overconsolidation ratio (OCR)

$$OCR = \frac{\sigma'_c}{\sigma'}$$

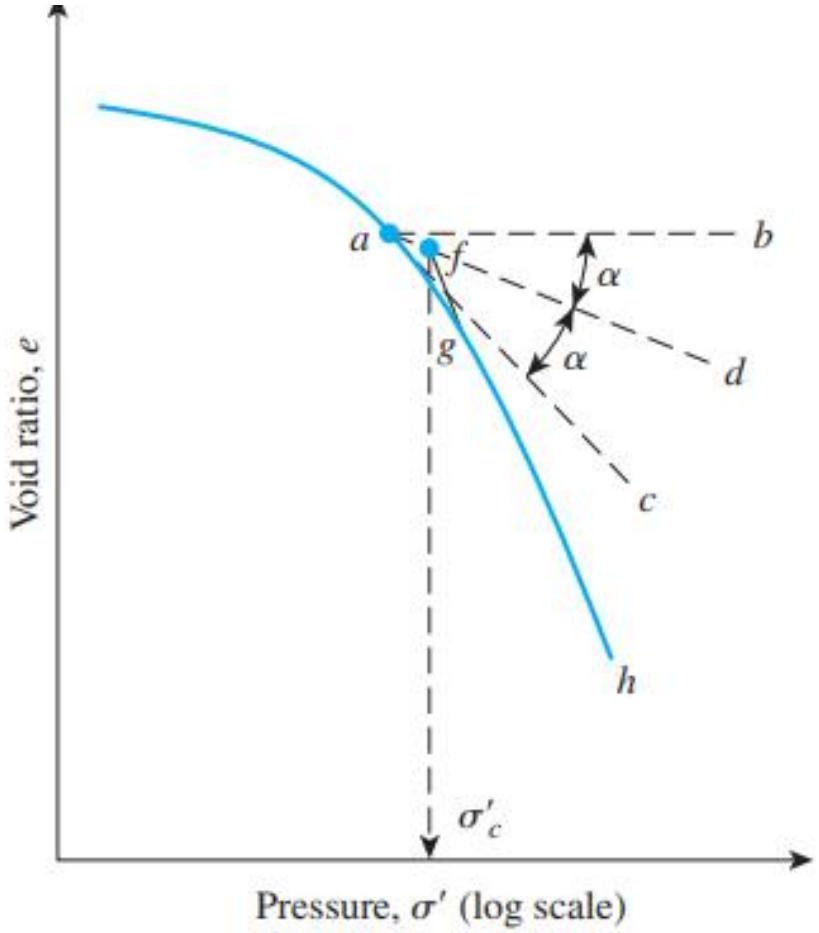



**Figure 11.13** Graphic procedure for determining preconsolidation pressure

# Effect of Disturbance on Void Ratio-Pressure Relationship

- A soil specimen will be remolded when it is subjected to some degree of disturbance.

**Normally Consolidated Clay of Low to Medium Plasticity**

1. In Figure 11.14, curve 2 is the laboratory e-log σ' plot. From this plot, determine the preconsolidation pressure ($\sigma'_c$) = ($\sigma'_o$) (that is, the present effective overburden pressure). Knowing where $\sigma'_c = \sigma'_o$, draw vertical line ab.
2. Calculate the void ratio in the field, $e_0$. Draw horizontal line cd.
3. Calculate $0.4e_o$ and draw line ef
4. Join points f and g, g is the point of intersection of lines ab and cd. This is the virgin compression curve.

If a soil is remolded completely, the general position of the e-log $\sigma\sigma'$ plot will be as represented by curve 3.

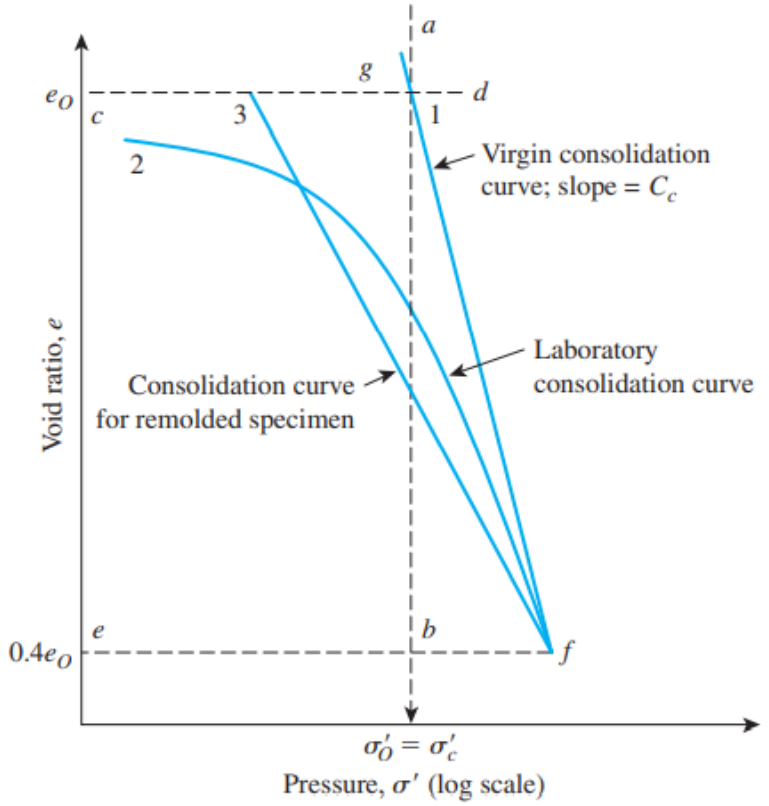



***Figure 11.14*** Consolidation characteristics of normally consolidated clay of low to medium sensitivity

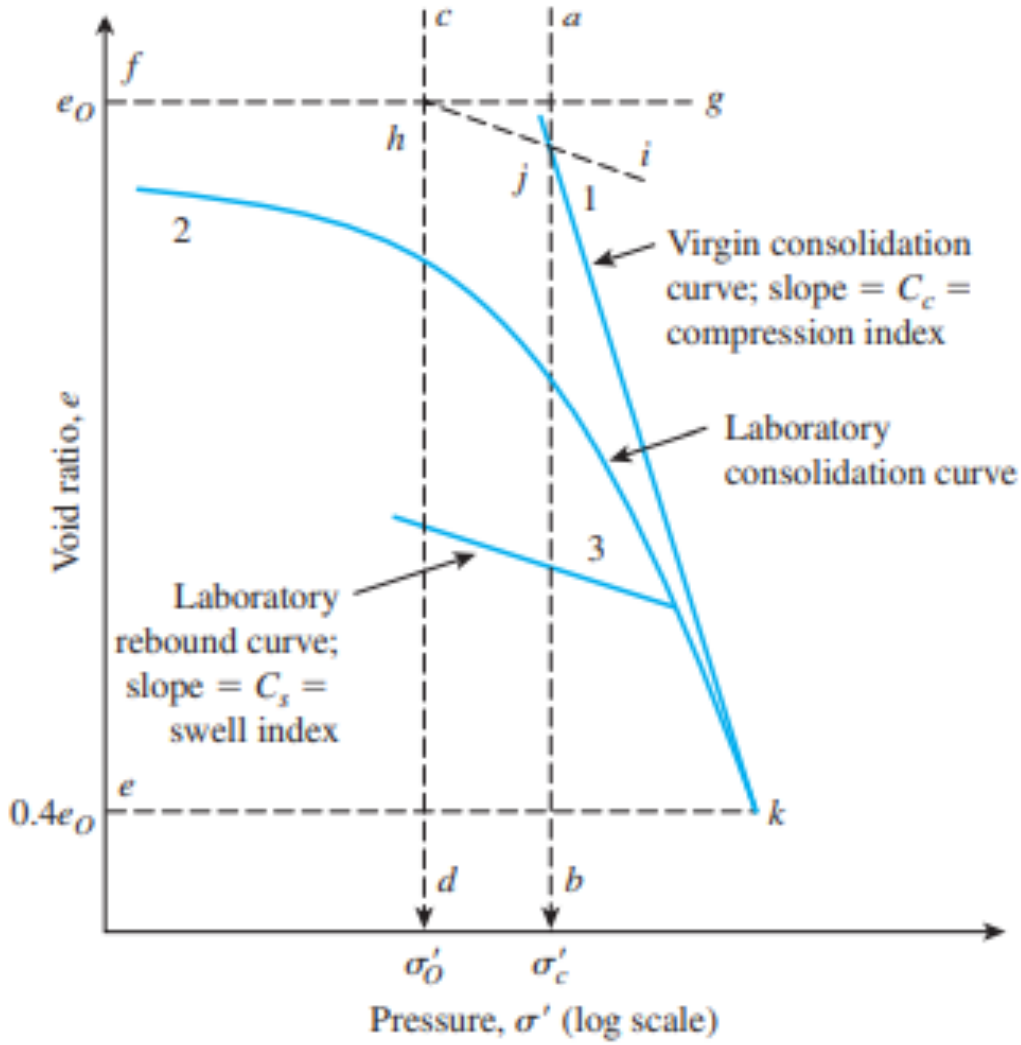
# Effect of Disturbance on Void Ratio-Pressure Relationship

**OverConsolidated Clay of Low to Medium Plasticity**

1. In Figure 11.15, curve 2 is the laboratory e-log σ’ plot(loading), and curve 3 is the laboratory unloading, or rebound, curve. From curve 2, determine the preconsolidation pressure $\sigma'_c$. Draw the vertical line ab.
2. Determine the field effective overburden pressure $\sigma'_o$. Draw vertical line cd.
3. Determine the void ratio in the field, $e_0$ . Draw the horizontal line fg. The point of intersection of lines fg and cd is h
4. Draw a line hi, which is parallel to curve 3. The point of intersection of line hi and ab is j.
5. Join points j and k. Point k is on curve 2, and its ordinate is $0.4e_0$.
6. The field consolidation plot will take a path hjk. The recompression path in the field is hj and is parallel to the laboratory rebound curve(Schmertmann, 1953)

# Effect of Disturbance on Void Ratio-Pressure Relationship

**OverConsolidated Clay of Low to Medium Plasticity**



# Calculation of Settlement from One-Dimensional Primary Consolidation

**Effective overburden pr. $\sigma'_0$, increased effective pr. $\Delta\sigma'$ → primary settlement $S_c$**

$$\Delta V = V_0 - V_1 = HA - (H - S_c)A = S_c A$$

$$\Delta V = S_c A = V_{v0} - V_{v1} = \Delta V_v$$

$$\Delta V_v = \Delta e V_s$$

$$V_s = \frac{V_0}{1 + e_0} = \frac{AH}{1 + e_0}$$

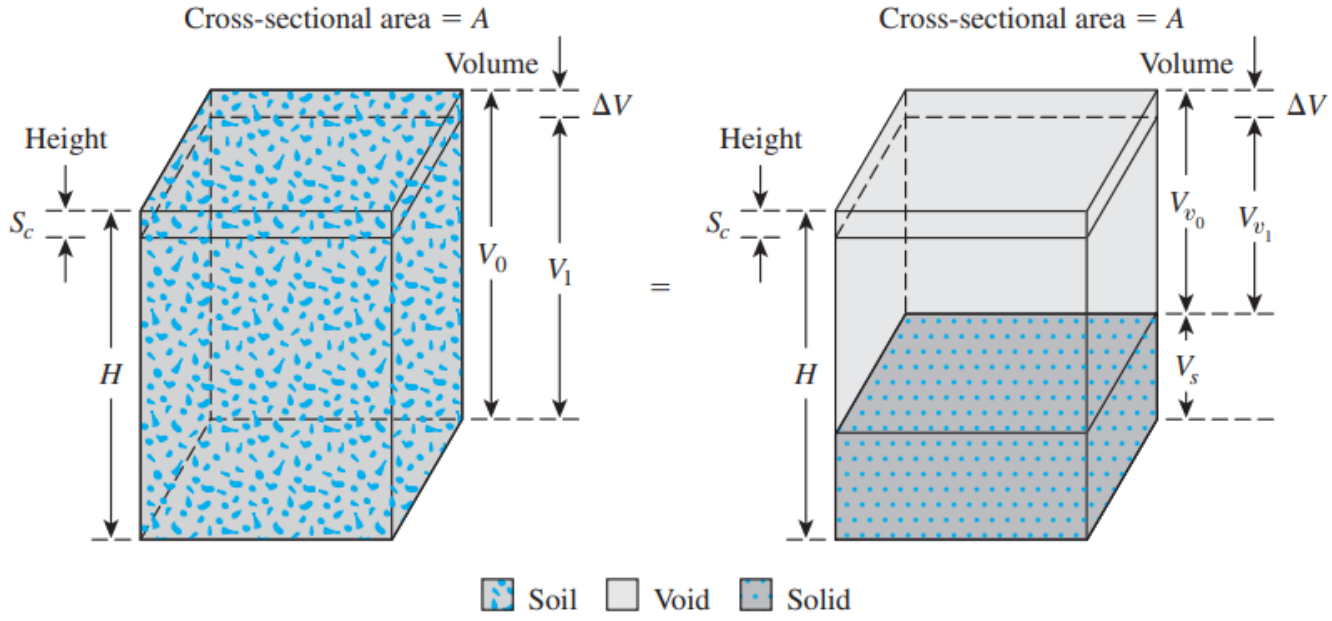



*Figure 11.16* Settlement caused by one-dimensional consolidation

# Calculation of Settlement from One-Dimensional Primary Consolidation

**Effective overburden pr. $\sigma'_0$, increased effective pr. $\Delta\sigma'$ → primary settlement $S_c$**

$$V_s = \frac{V_0}{1+e_0} = \frac{AH}{1+e_0}$$

$$\Delta V = S_c A = \Delta e V_s = \frac{AH}{1+e_0}\Delta e$$

$$S_c = H\left(\frac{\Delta e}{1+e_0}\right)$$

- For normally consolidated clays

$$\Delta e = C_c[\log(\sigma'_0 + \Delta\sigma') - log\sigma'_0]$$

$C_c$ = Compression index

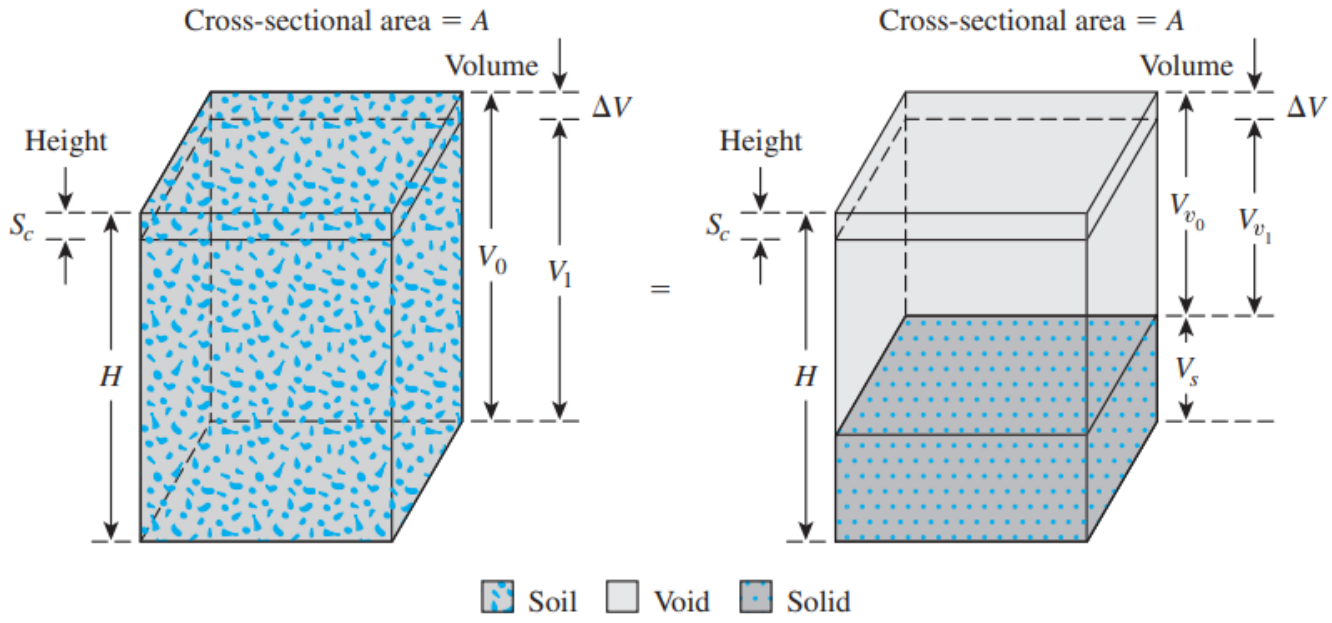



*Figure 11.16* Settlement caused by one-dimensional consolidation

# Calculation of Settlement from One-Dimensional Primary Consolidation

**Effective overburden pr. $\sigma'_0$, increased effective pr. $\Delta\sigma'$ → primary settlement $S_c$**

$$S_c = \frac{C_c H}{(1+e_0)} \log\left(\frac{\sigma'_0 + \Delta\sigma'}{\sigma'_0}\right)$$

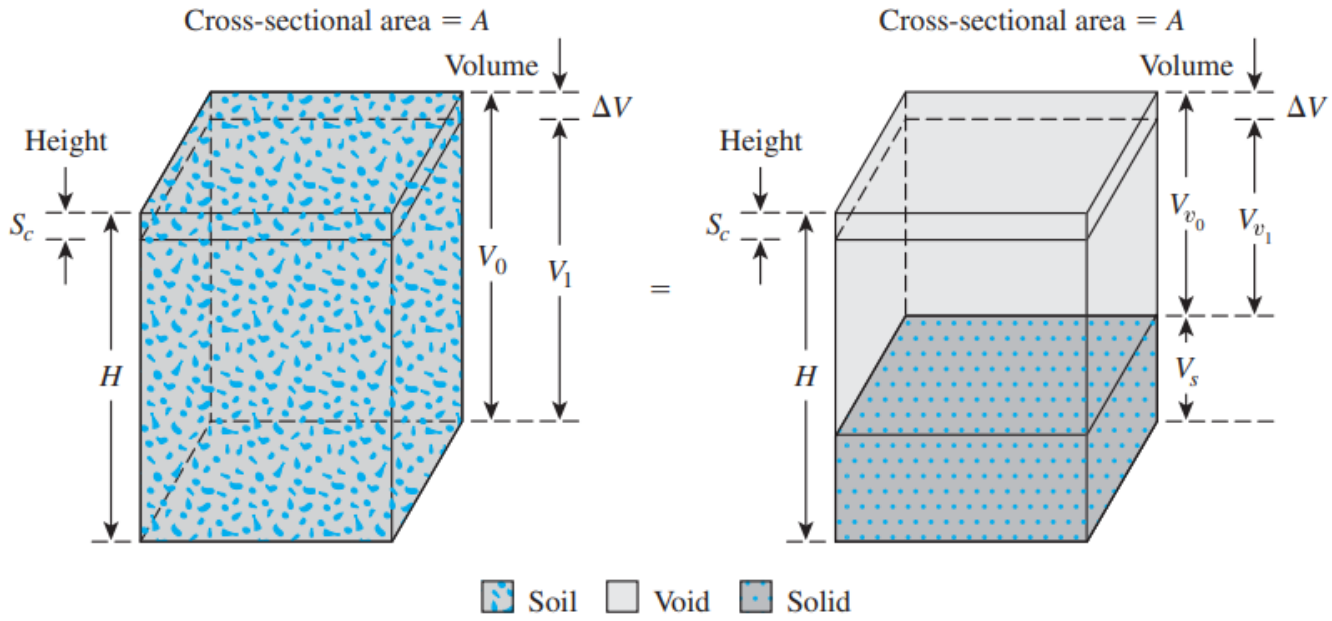



*Figure 11.16* Settlement caused by one-dimensional consolidation

# Calculation of Settlement from One-Dimensional Primary Consolidation

**Effective overburden pr. $\sigma'_0$, increased effective pr. $\Delta\sigma'$ → primary settlement $S_c$**

- In overconsolidated clays, for

$$\sigma'_0 + \Delta\sigma' \leq \sigma'_c$$

$$\Delta e = C_s[\log(\sigma'_0 + \Delta\sigma') - log\sigma'_0]$$

$$S_c = \frac{C_s H}{(1+e_0)} \log\left(\frac{\sigma'_0 + \Delta\sigma'}{\sigma'_0}\right)$$

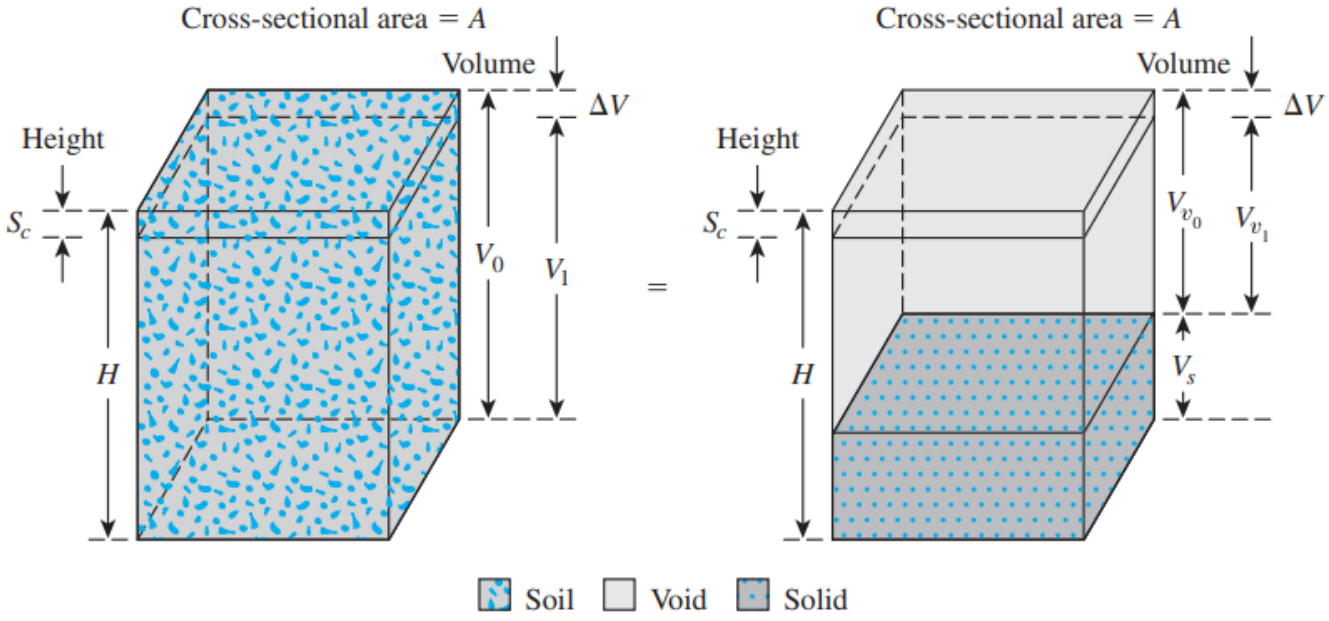



*Figure 11.16* Settlement caused by one-dimensional consolidation

# Calculation of Settlement from One-Dimensional Primary Consolidation

**Effective overburden pr. $\sigma'_0$, increased effective pr. $\Delta\sigma'$ → primary settlement $S_c$**

- In overconsolidated clays, for

$$\sigma'_0 + \Delta\sigma' \geq \sigma'_c$$

$$S_c = \frac{C_s H}{(1+e_0)} \log\left(\frac{\sigma'_c}{\sigma'_0}\right) + \frac{C_c H}{(1+e_0)} \log\left(\frac{\sigma'_0 + \Delta\sigma'}{\sigma'_c}\right)$$

- However, if the e-log p curve is given, it is possible to simply to pick Δe of the plot for the appropriate range of pressures

# Compression Index ($C_c$)

- Terzaghi and Peck (1967)
  - For undisturbed clays :

$$C_c = 0.009(LL - 10)$$

  - For remolded clays

$$C_c = 0.007(LL - 10)$$

- Rendon-Herrero (1983)

$$C_c = 0.141 G_s^{1.2} \left( \frac{1 + e_0}{G_s} \right)^{2.38}$$

- Nagaraj and Murty (1985)

$$C_c = 0.2343 \left[ \frac{LL(\%)}{100} \right] G_s$$

# Compression Index ($C_c$)

**Table 11.6** Correlations for Compression Index, $C_c$*

| Equation | Reference | Region of applicability |
|---|---|---|
| $C_c = 0.007(LL - 7)$ | Skempton (1944) | Remolded clays |
| $C_c = 0.01 w_N$ | | Chicago clays |
| $C_c = 1.15(e_O - 0.27)$ | Nishida (1956) | All clays |
| $C_c = 0.30(e_O - 0.27)$ | Hough (1957) | Inorganic cohesive soil: silt, silty clay, clay |
| $C_c = 0.0115 w_N$ | | Organic soils, peats, organic silt, and clay |
| $C_c = 0.0046(LL - 9)$ | | Brazilian clays |
| $C_c = 0.75(e_O - 0.5)$ | | Soils with low plasticity |
| $C_c = 0.208 e_O + 0.0083$ | | Chicago clays |
| $C_c = 0.156 e_O + 0.0107$ | | All clays |

**After Rendon-Herrero, 1980. With permission from ASCE.*

*Note:* $e_O$ = *in situ* void ratio; $w_N$ = *in situ* water content.

# Swelling Index ($C_s$)

$$C_s \approx \frac{1}{5} \; to \frac{1}{10} \; C_c$$

- Nagaraj and Murty (1985 )

$$C_s = 0.0463 \left[\frac{LL(\%)}{100}\right] G_s$$

*Table 11.7* Compression and Swell of Natural Soils

| Soil | Liquid limit | Plastic limit | Compression index, $C_c$ | Swell index, $C_s$ |
|---|---|---|---|---|
| Boston blue clay | 41 | 20 | 0.35 | 0.07 |
| Chicago clay | 60 | 20 | 0.4 | 0.07 |
| Ft. Gordon clay, Georgia | 51 | 26 | 0.12 | — |
| New Orleans clay | 80 | 25 | 0.3 | 0.05 |
| Montana clay | 60 | 28 | 0.21 | 0.05 |

# Time Rate of Consolidation

- Terzaghi(1925)
- The mathematical derivations are based on the following six assumptions (also see Taylor, 1948):

1. The clay-water system is homogeneous.
2. Saturation is complete.
3. Compressibility of water is negligible.
4. Compressibility of soil grains is negligible (but soil grains rearrange).
5. The flow of water is in one direction only (that is, in the direction of compression.)
6. Darcy's law is valid.

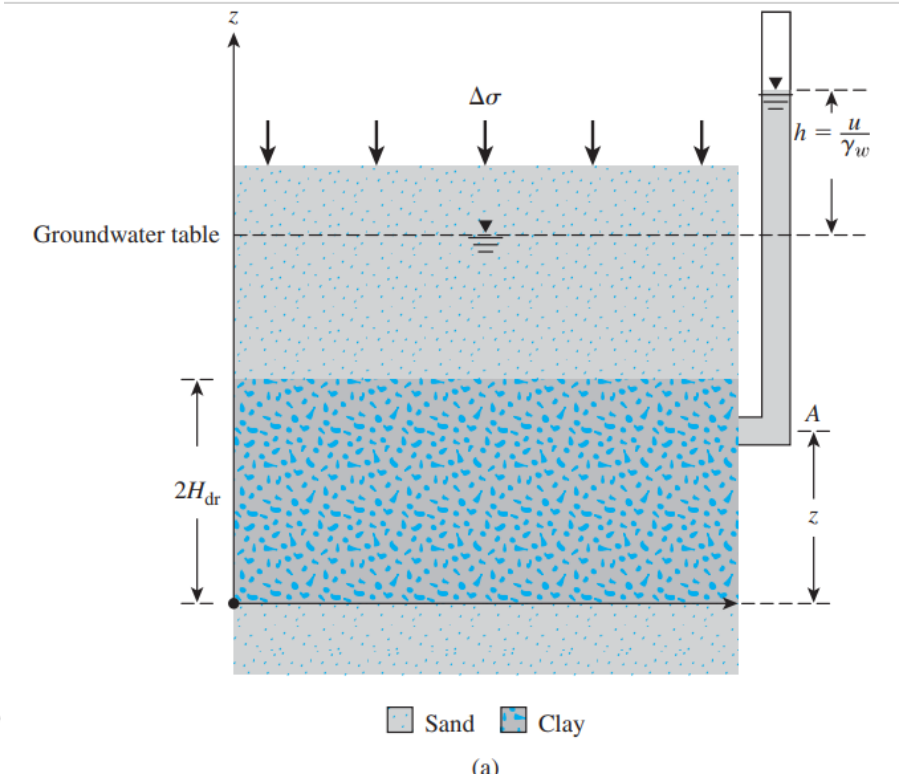
# Time Rate of Consolidation

$$\begin{pmatrix} Rate\ of\ outflow \\ of\ water \end{pmatrix} - \begin{pmatrix} Rate\ of\ inflow \\ of\ water \end{pmatrix} =$$

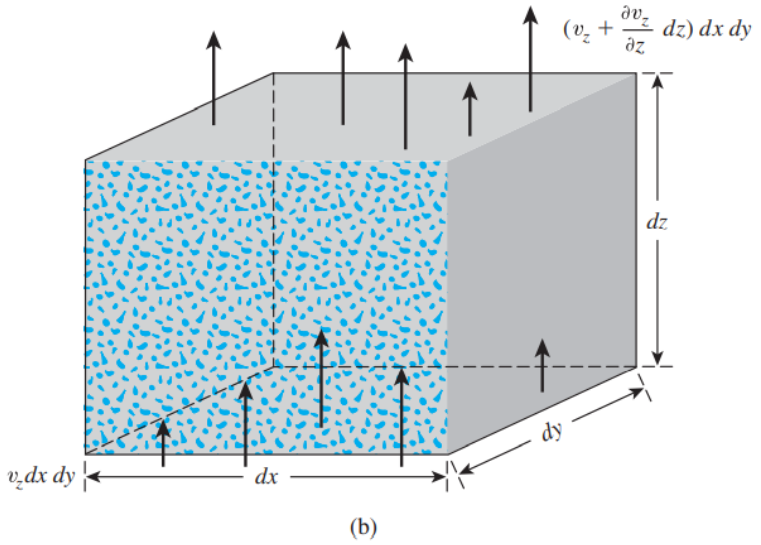
$$(Rate\ of volume\ change)$$

$$\left( v_z + \frac{\partial v_z}{\partial z} dz \right) dx\, dy - v_z\, dx\, dy = \frac{\partial V}{\partial t}$$

$$\boxed{\frac{\partial v_z}{\partial z} dx\, dy\, dz = \frac{\partial V}{\partial t}}$$




(a)




(b)

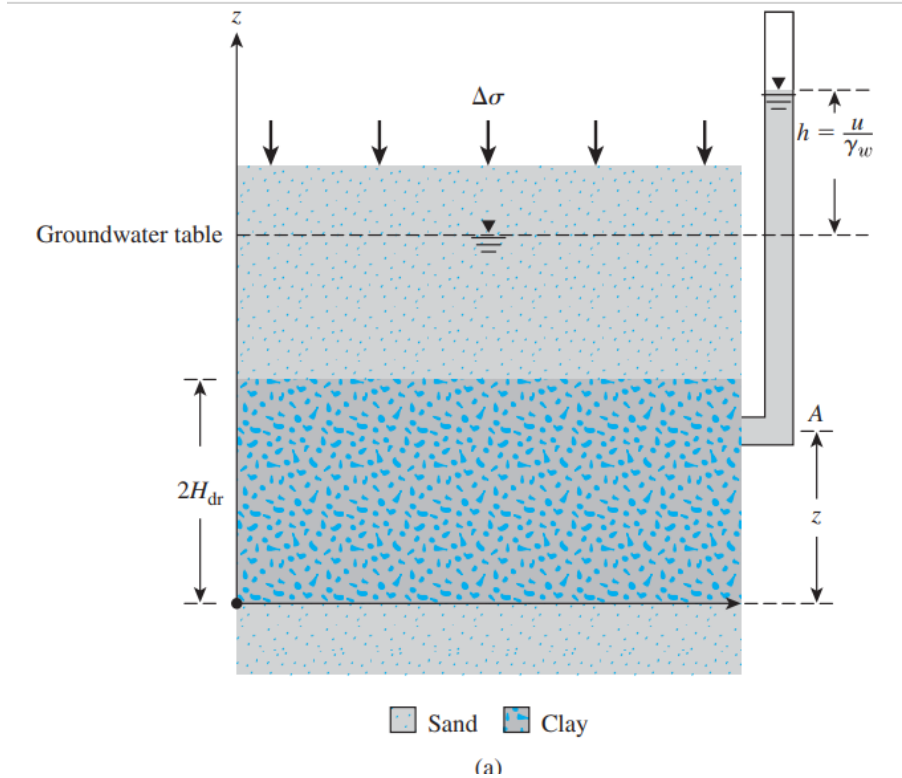
# Time Rate of Consolidation

- Using Darcy's law

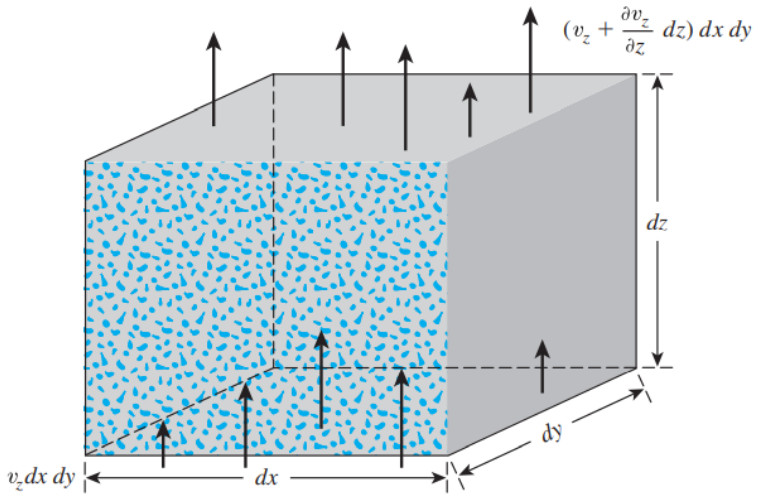
$$v_z = ki = -k\frac{\partial h}{\partial z} = -\frac{k}{\gamma_w}\frac{\partial u}{\partial z}$$

- Elevation head constant
- u = excess pore water pressure caused by the increase of stress

$$-\frac{k}{\gamma_w}\frac{\partial^2 u}{\partial z^2} = \frac{1}{dx\,dy\,dz}\frac{\partial V}{\partial t}$$



(a)



(b)

# Time Rate of Consolidation

- During consolidaiton

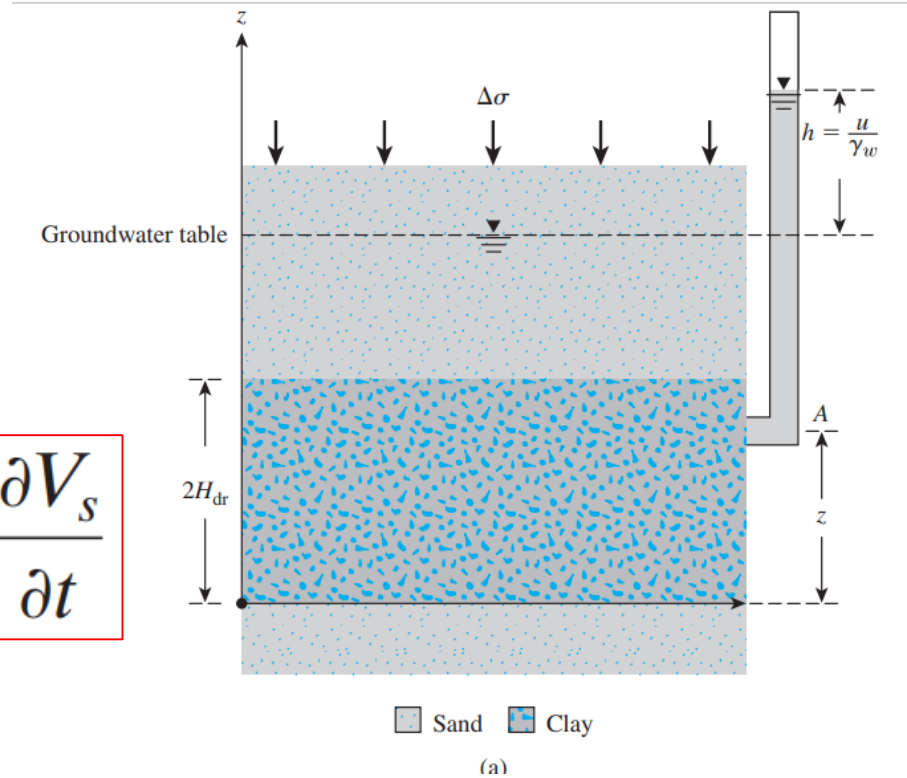
$$\frac{\partial V}{\partial t} = \frac{\partial V_v}{\partial t} = \frac{\partial (V_s + eV_s)}{\partial t} = \frac{\partial V_s}{\partial t} + V_s \frac{\partial e}{\partial t} + e \frac{\partial V_s}{\partial t}$$

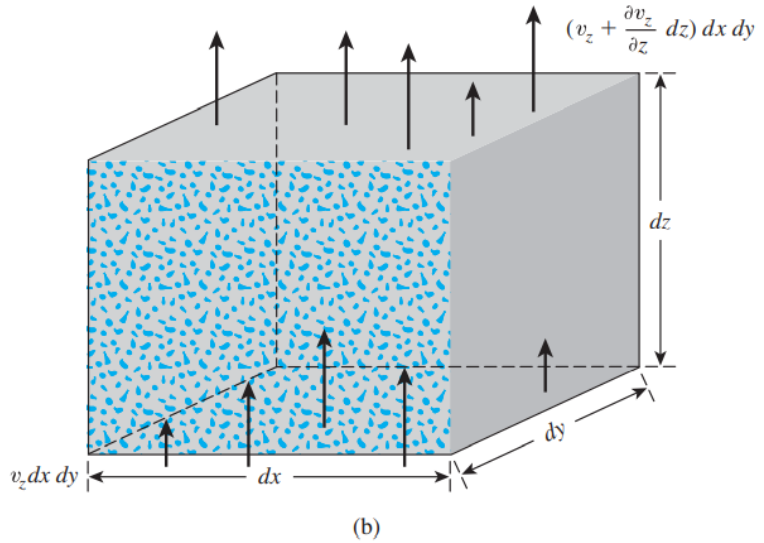
- Assuming that soil solids are incompressible

$$V_s = \frac{V}{1 + e_o} = \frac{dx\, dy\, dz}{1 + e_o}$$

- Substitution for $\partial V_s / \partial t$ and $V_s$ in yields

$$\frac{\partial V}{\partial t} = \frac{dx\, dy\, dz}{1 + e_o} \frac{\partial e}{\partial t}$$



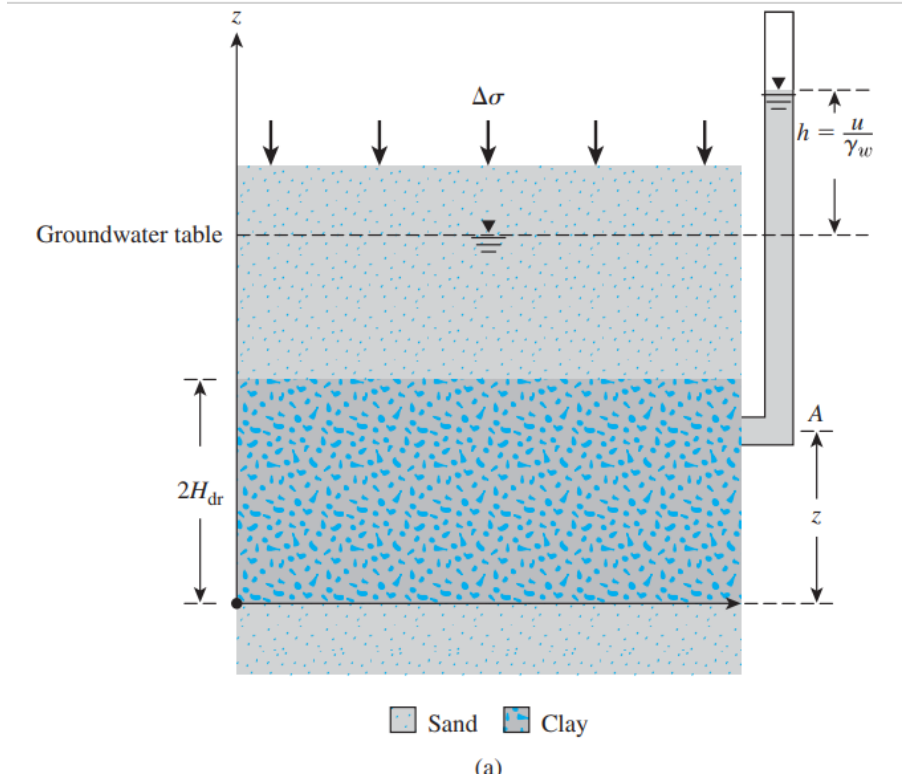



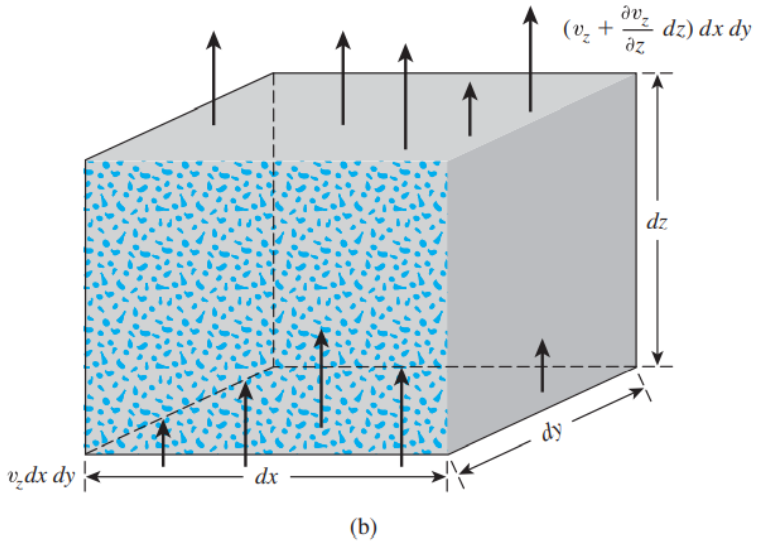
# Time Rate of Consolidation

- Combining the Eqs. Gives

$$-\frac{k}{\gamma_w}\frac{\partial^2 u}{\partial z^2} = \frac{1}{1+e_o}\frac{\partial e}{\partial t}$$

- The change in the void ratio is caused by the increase of effective stress (that is, decrease of excess pore water pressure).



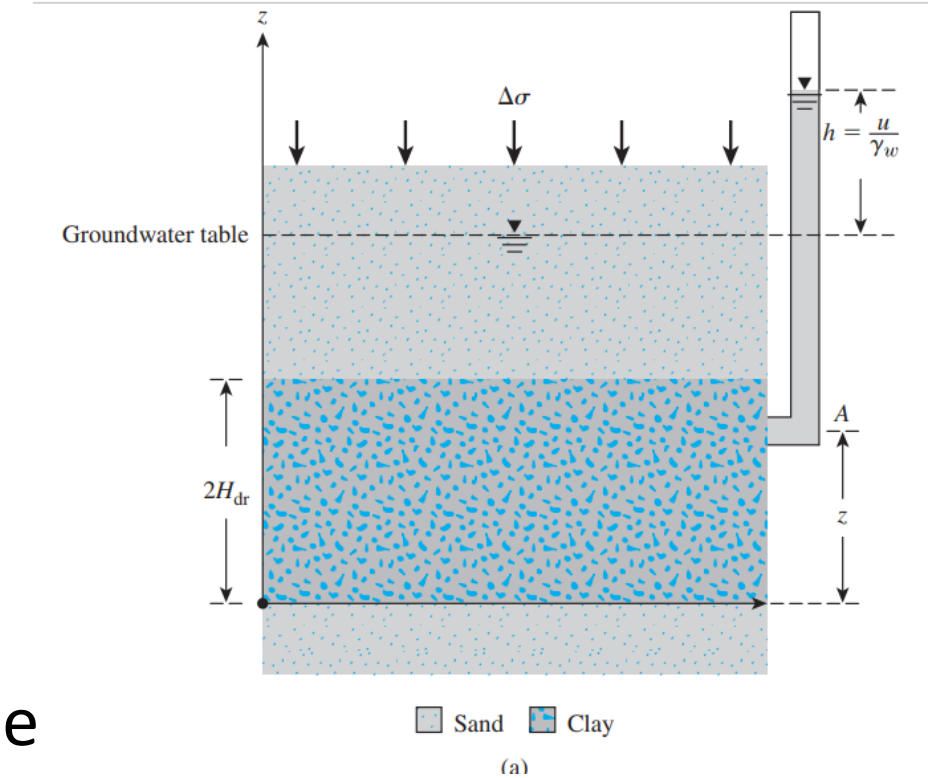



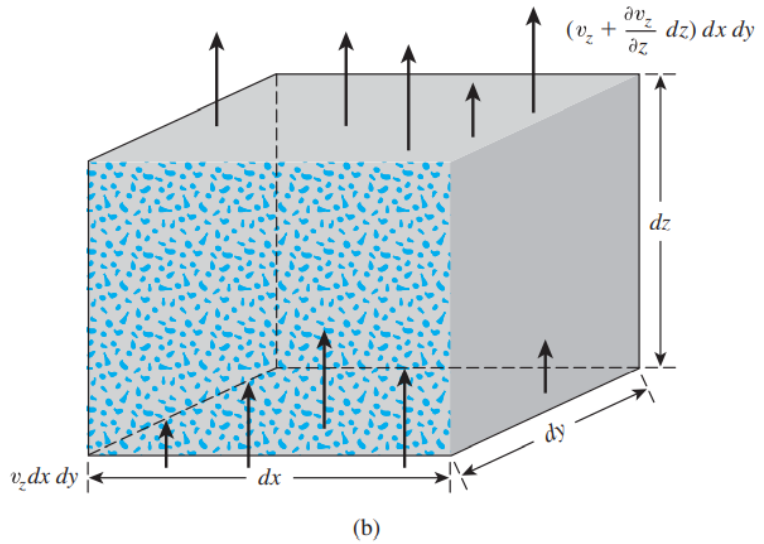
# Time Rate of Consolidation

- Assuming that they are linearly related, we have

$$\partial e = a_v \partial(\Delta\sigma') = -a_v \partial u$$

- $a_v$ = coefficient of compressibility ($a_v$ can be considered to be constant for a narrow range of pressure increase)



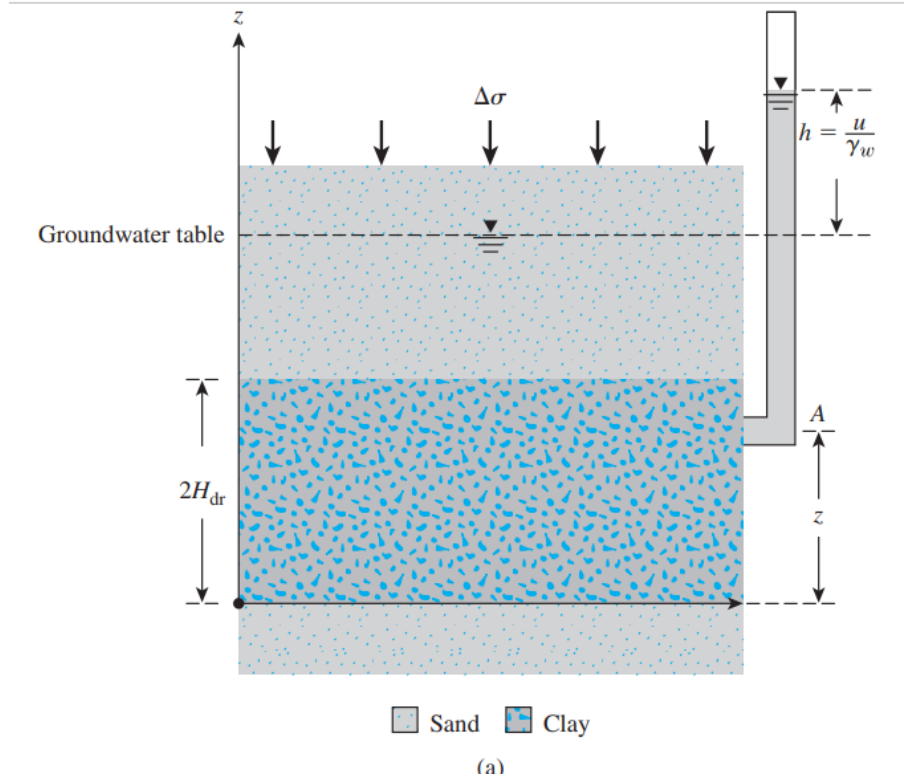



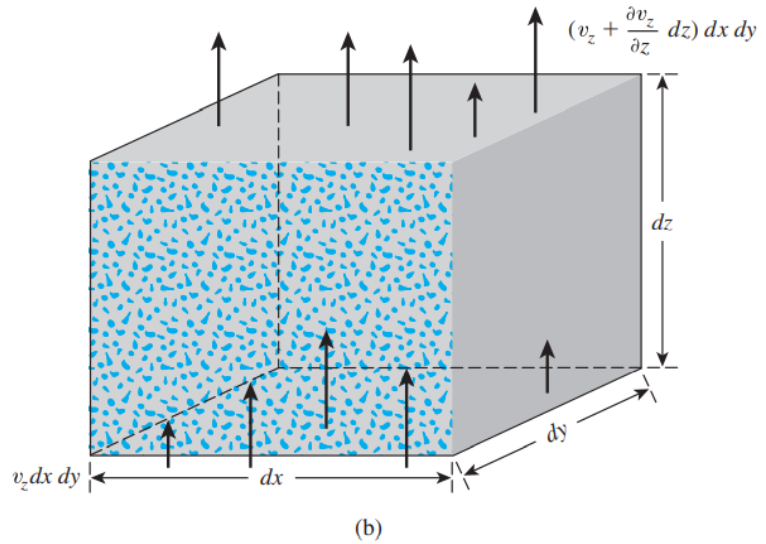
# Time Rate of Consolidation

- Combining the Eqs. gives

$$-\frac{k}{\gamma_w}\frac{\partial^2 u}{\partial z^2} = -\frac{a_v}{1+e_o}\frac{\partial u}{\partial t} = -m_v\frac{\partial u}{\partial t}$$

- $m_v$ = coefficient of volume compressibility
- $= a_v/(1+e_0$



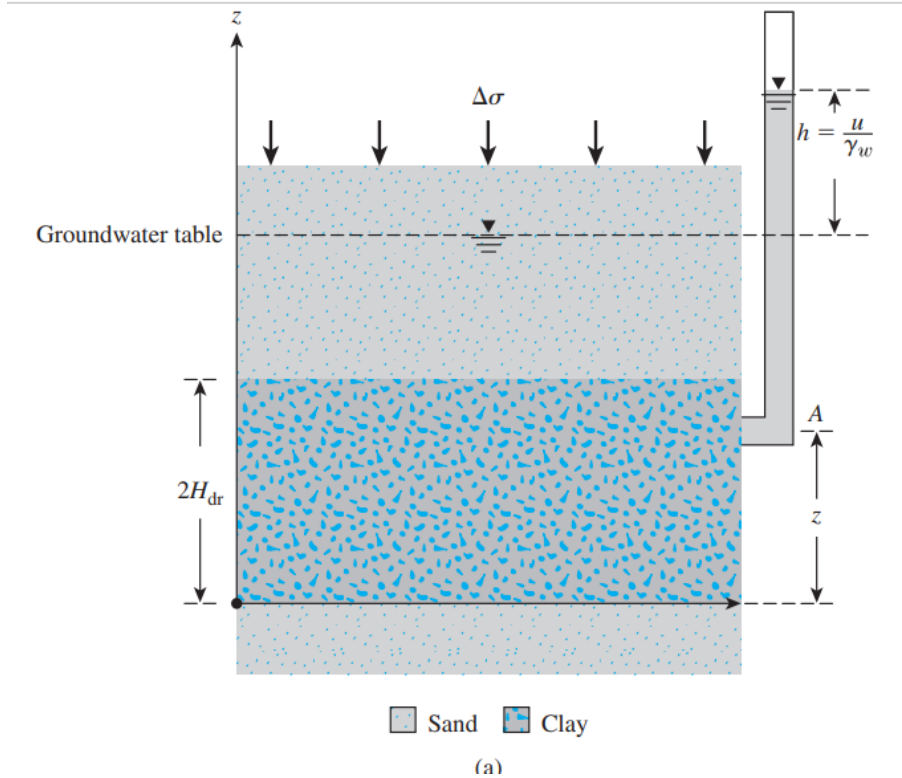



# Time Rate of Consolidation

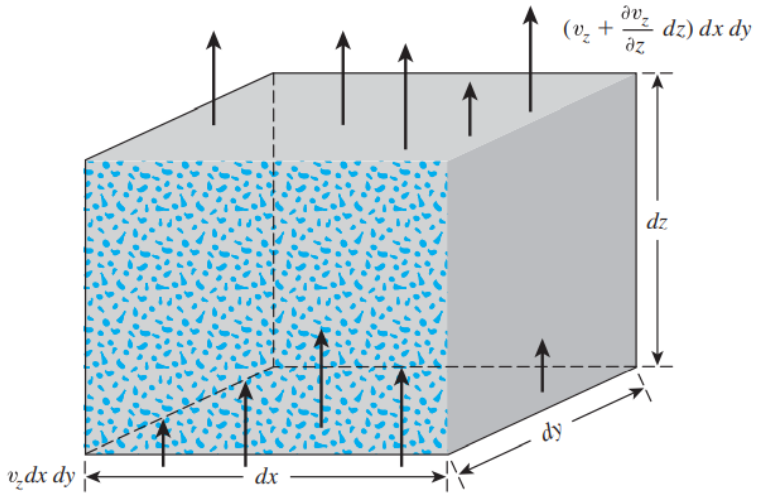
$$\frac{\partial u}{\partial t} = c_v \frac{\partial^2 u}{\partial z^2}$$

- $c_v$ = coefficient of consolidation = $\dfrac{k}{\gamma_w m_v}$
-

$$c_v = \frac{k}{\gamma_w m_v} = \frac{k}{\gamma_w \left( \dfrac{a_v}{1 + e_o} \right)}$$



(a)

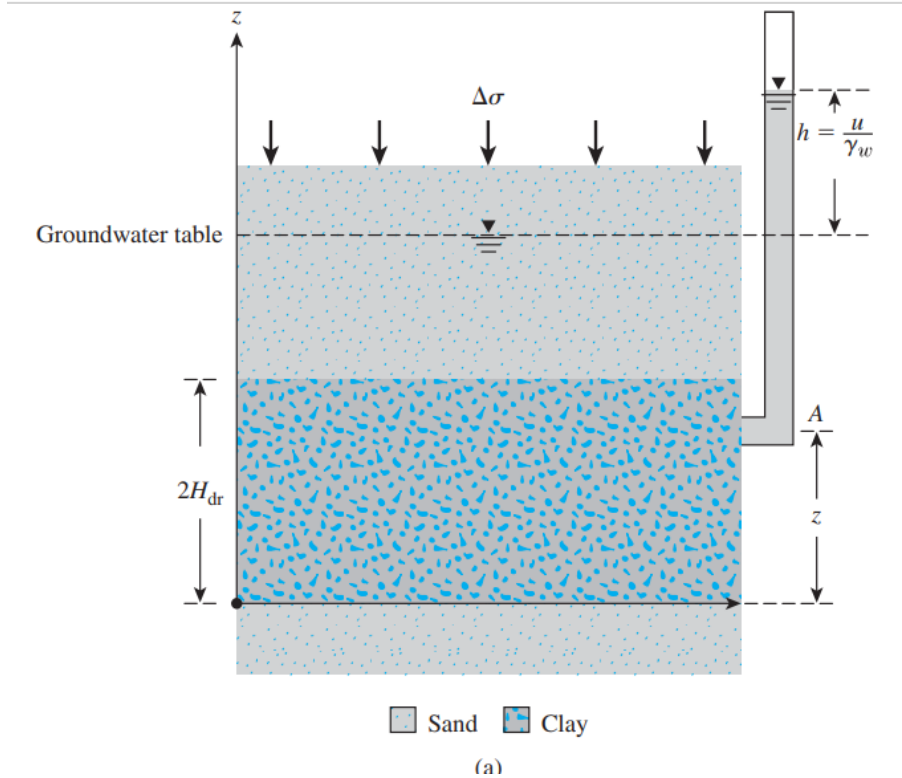


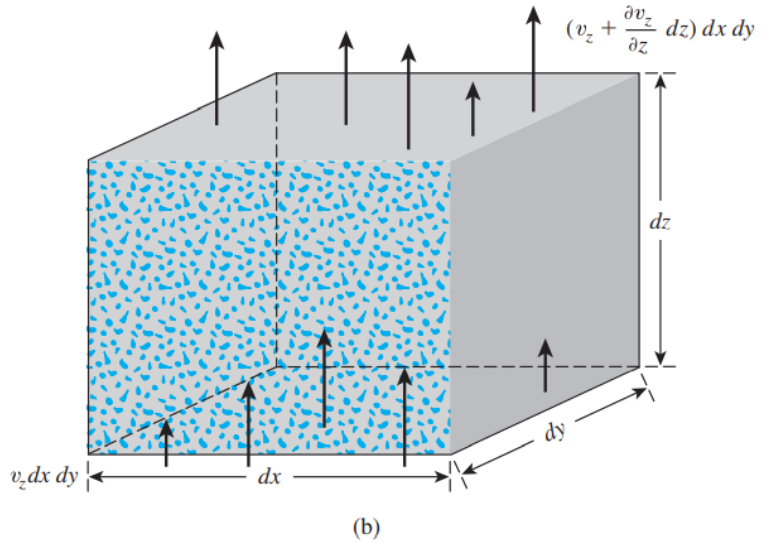
(b)

# Time Rate of Consolidation

- *Boundary conditions:*
- $Z=0, u=0$
- $Z=2H_{dr}, u=0$
- $t=0, u=u_0$
- The solution yields

$$u = \sum_{m=0}^{m=\infty} \left[ \frac{2u_o}{M} \sin\left( \frac{Mz}{H_{\text{dr}}} \right) \right] e^{-M^2 T_v}$$



# Time Rate of Consolidation

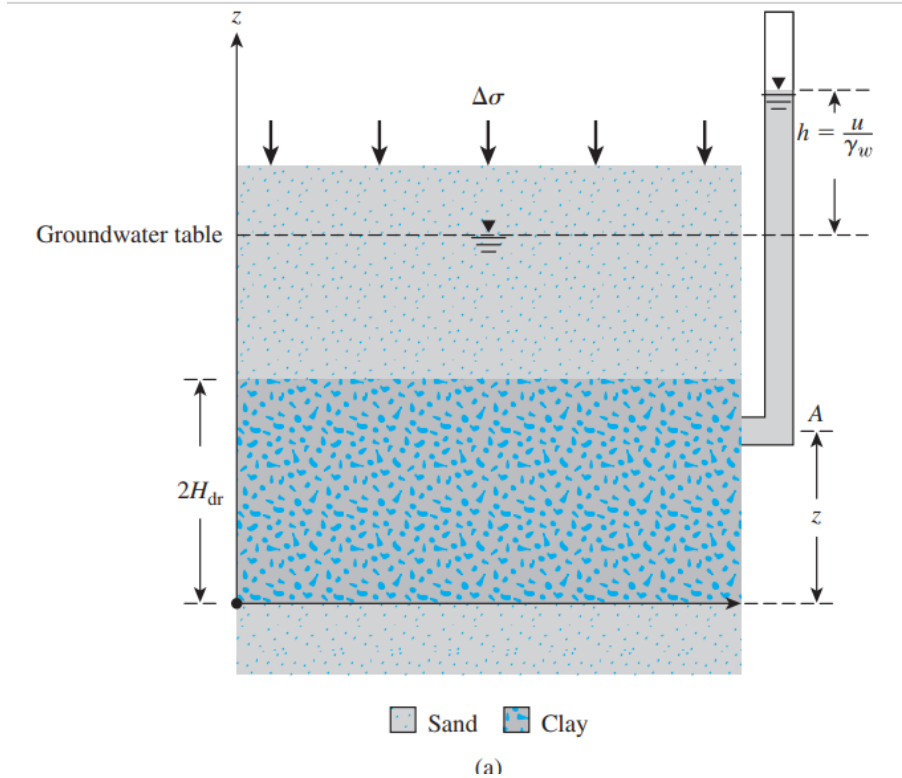
$$u = \sum_{m=0}^{m=\infty} \left[ \frac{2u_o}{M} \sin\left( \frac{Mz}{H_{\text{dr}}} \right) \right] e^{-M^2 T_v}$$

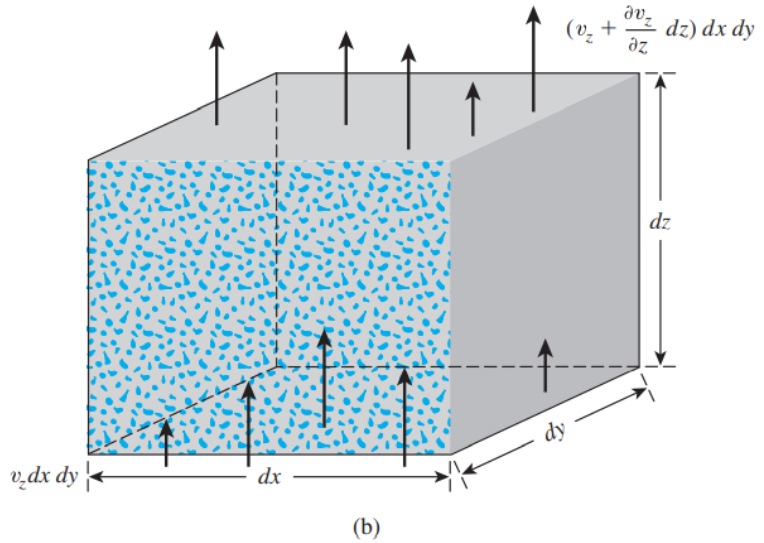
- Where , m is an integer

$$M = \left( \frac{\pi}{2} \right) (2m + 1)$$

$u_0 =$ initial excess pore water pressure

$$T_v = \frac{c_v t}{H_{\text{dr}}^2} = \text{time factor}$$



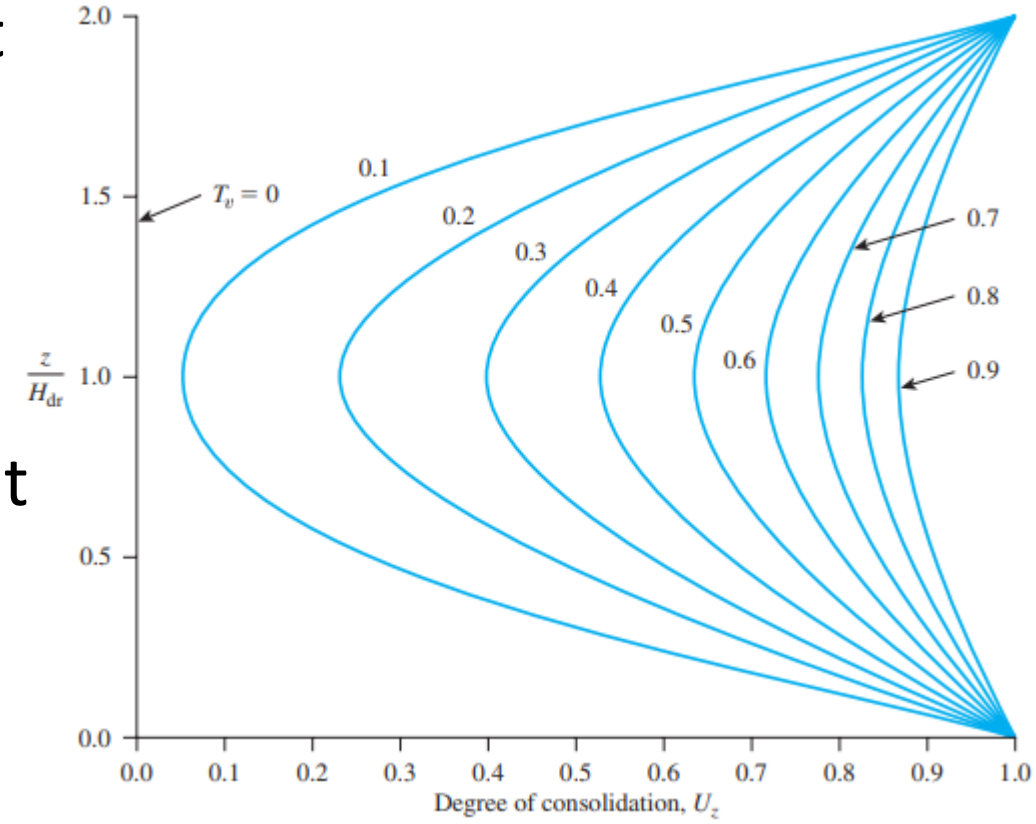
# Time Rate of Consolidation

- Degree of consolidation at a distance z at any time t

$$U_z = \frac{u_o - u_z}{u_o} = 1 - \frac{u_z}{u_o}$$

- $U_z$ = excess pore water pressure at time t
- average degree of consolidation for the entire depth of the clay layer

$$U = \frac{S_{c(t)}}{S_c} = 1 - \frac{\left(\frac{1}{2H_{dr}}\right)\int_0^{2H_{dr}} u_z\, dz}{u_o}$$



# Time Rate of Consolidation

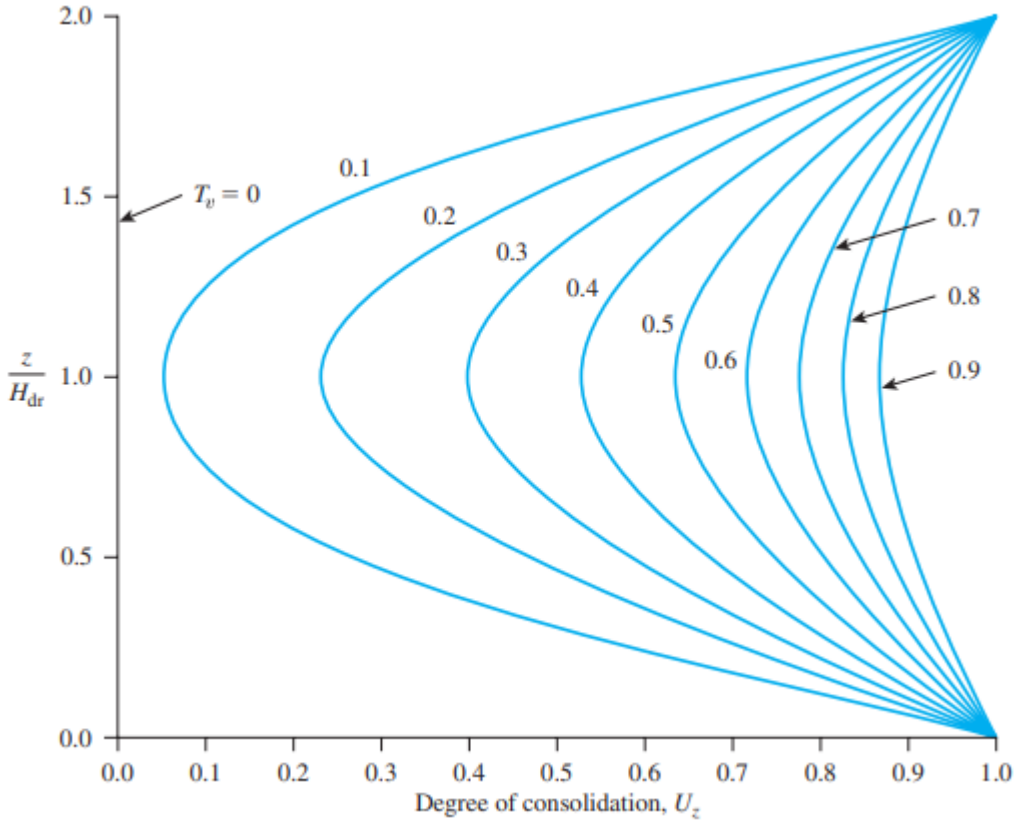
$$U = \frac{S_{c(t)}}{S_c} = 1 - \frac{\left(\frac{1}{2H_{dr}}\right)\int_0^{2H_{dr}} u_z\, dz}{u_o}$$

- Where U = average degree of consolidation

  $S_{c(t)}$ = settlement of the layer at time t

  $S_c$ = ultimate settlement of the layer from primary consolidation

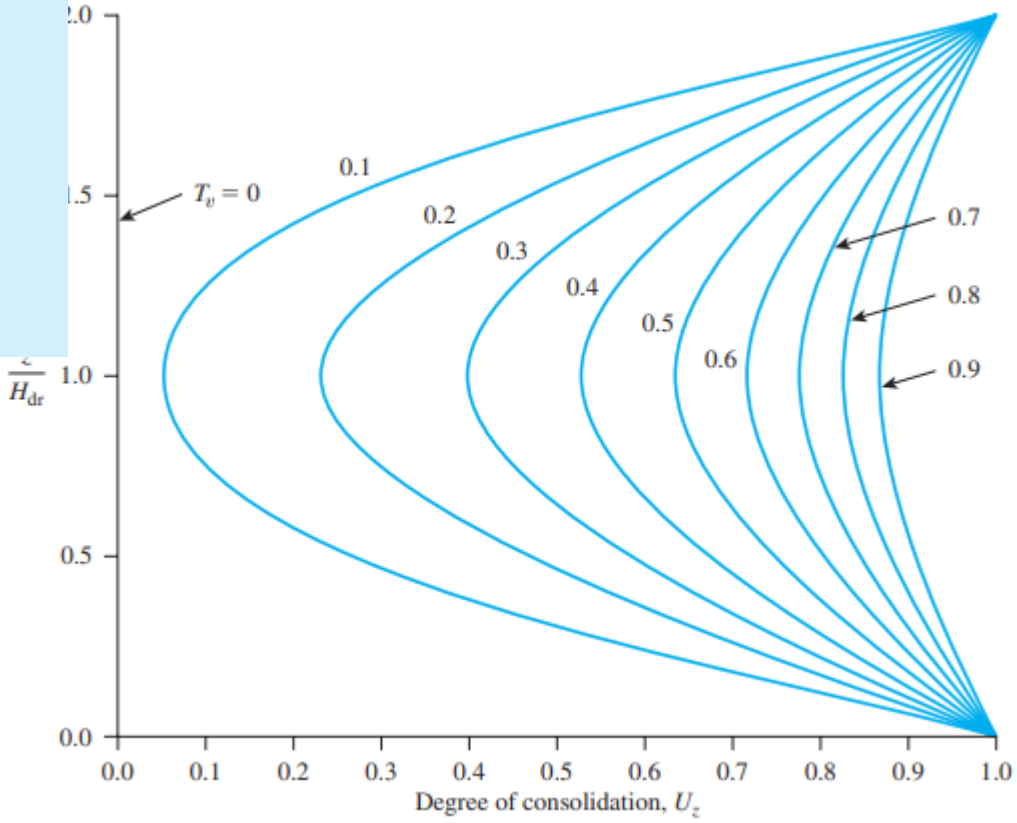
$$U = 1 - \sum_{m=0}^{m=\infty} \frac{2}{M^2} e^{-M^2 T_v}$$



# Time Rate of Consolidation

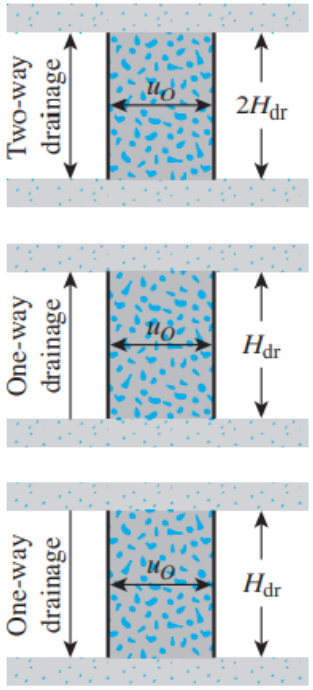
$$\text{For } U = 0 \text{ to } 60\%, \quad T_v = \frac{\pi}{4}\left(\frac{U\%}{100}\right)^2$$

$$\text{For } U > 60\%, \quad T_v = 1.781 - 0.933 \log(100 - U\%)$$



# Time Rate of Consolidation

- Variation of average degree of consolidation with time factor, Tv ($u_0$ constant with depth)




Different types of drainage with $u_O$ constant

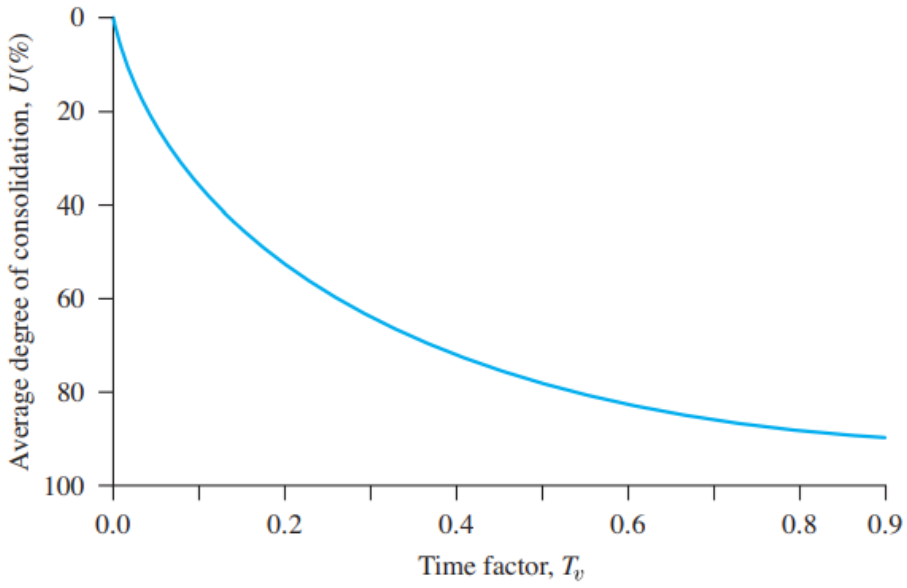



***Figure 11.24*** Variation of average degree of consolidation with time factor, $T_v$ ($u_O$ constant with depth)

# Time Rate of Consolidation

**Table 11.8** Variation of $T_v$ with $U$

| $U$ (%) | $T_v$ | $U$ (%) | $T_v$ | $U$ (%) | $T_v$ | $U$ (%) | $T_v$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 26 | 0.0531 | 52 | 0.212 | 78 | 0.529 |
| 1 | 0.00008 | 27 | 0.0572 | 53 | 0.221 | 79 | 0.547 |
| 2 | 0.0003 | 28 | 0.0615 | 54 | 0.230 | 80 | 0.567 |
| 3 | 0.00071 | 29 | 0.0660 | 55 | 0.239 | 81 | 0.588 |
| 4 | 0.00126 | 30 | 0.0707 | 56 | 0.248 | 82 | 0.610 |
| 5 | 0.00196 | 31 | 0.0754 | 57 | 0.257 | 83 | 0.633 |
| 6 | 0.00283 | 32 | 0.0803 | 58 | 0.267 | 84 | 0.658 |
| 7 | 0.00385 | 33 | 0.0855 | 59 | 0.276 | 85 | 0.684 |
| 8 | 0.00502 | 34 | 0.0907 | 60 | 0.286 | 86 | 0.712 |
| 9 | 0.00636 | 35 | 0.0962 | 61 | 0.297 | 87 | 0.742 |
| 10 | 0.00785 | 36 | 0.102 | 62 | 0.307 | 88 | 0.774 |
| 11 | 0.0095 | 37 | 0.107 | 63 | 0.318 | 89 | 0.809 |
| 12 | 0.0113 | 38 | 0.113 | 64 | 0.329 | 90 | 0.848 |
| 13 | 0.0133 | 39 | 0.119 | 65 | 0.304 | 91 | 0.891 |
| 14 | 0.0154 | 40 | 0.126 | 66 | 0.352 | 92 | 0.938 |
| 15 | 0.0177 | 41 | 0.132 | 67 | 0.364 | 93 | 0.993 |
| 16 | 0.0201 | 42 | 0.138 | 68 | 0.377 | 94 | 1.055 |
| 17 | 0.0227 | 43 | 0.145 | 69 | 0.390 | 95 | 1.129 |
| 18 | 0.0254 | 44 | 0.152 | 70 | 0.403 | 96 | 1.219 |
| 19 | 0.0283 | 45 | 0.159 | 71 | 0.417 | 97 | 1.336 |
| 20 | 0.0314 | 46 | 0.166 | 72 | 0.431 | 98 | 1.500 |
| 21 | 0.0346 | 47 | 0.173 | 73 | 0.446 | 99 | 1.781 |
| 22 | 0.0380 | 48 | 0.181 | 74 | 0.461 | 100 | $\infty$ |
| 23 | 0.0415 | 49 | 0.188 | 75 | 0.477 | | |
| 24 | 0.0452 | 50 | 0.197 | 76 | 0.493 | | |
| 25 | 0.0491 | 51 | 0.204 | 77 | 0.511 | | |

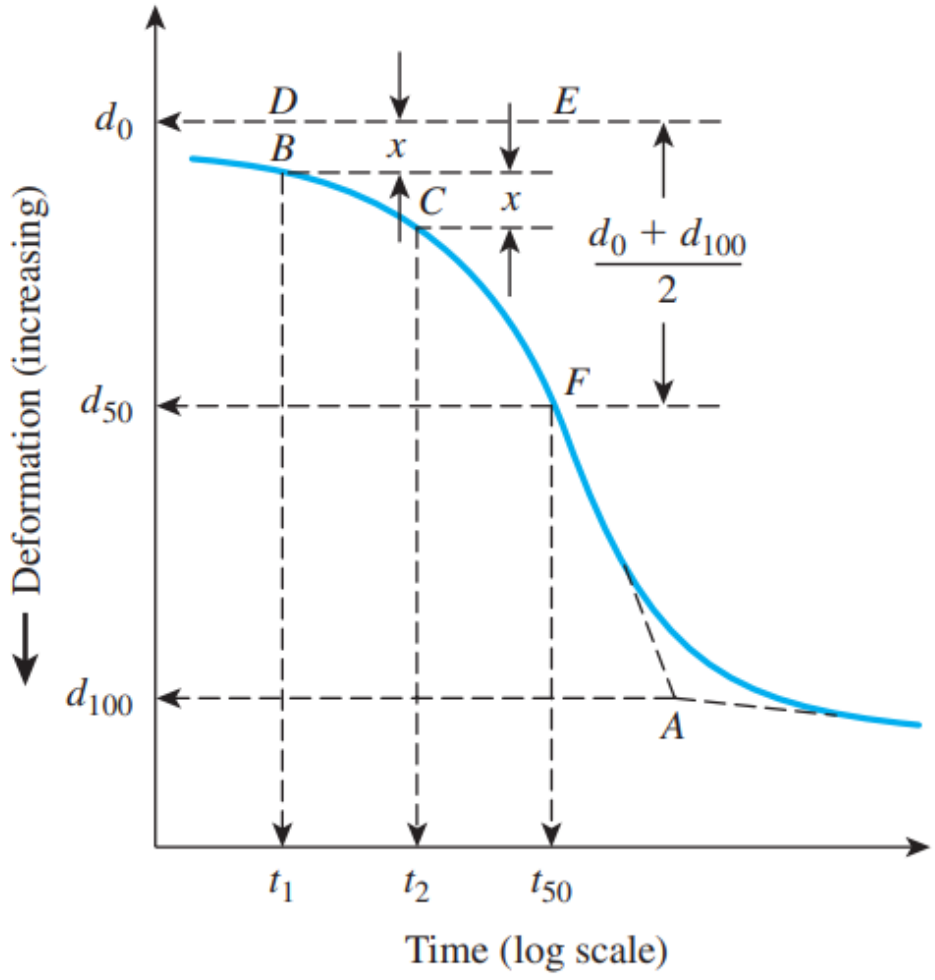
# Coefficient of Consolidaiton

- logarithm-of-time method proposed by Casagrande and Fandum(1940)
- square-root-of-time method suggested by Taylor(1942)

# Logarithm-of-time method

- Construction procedure

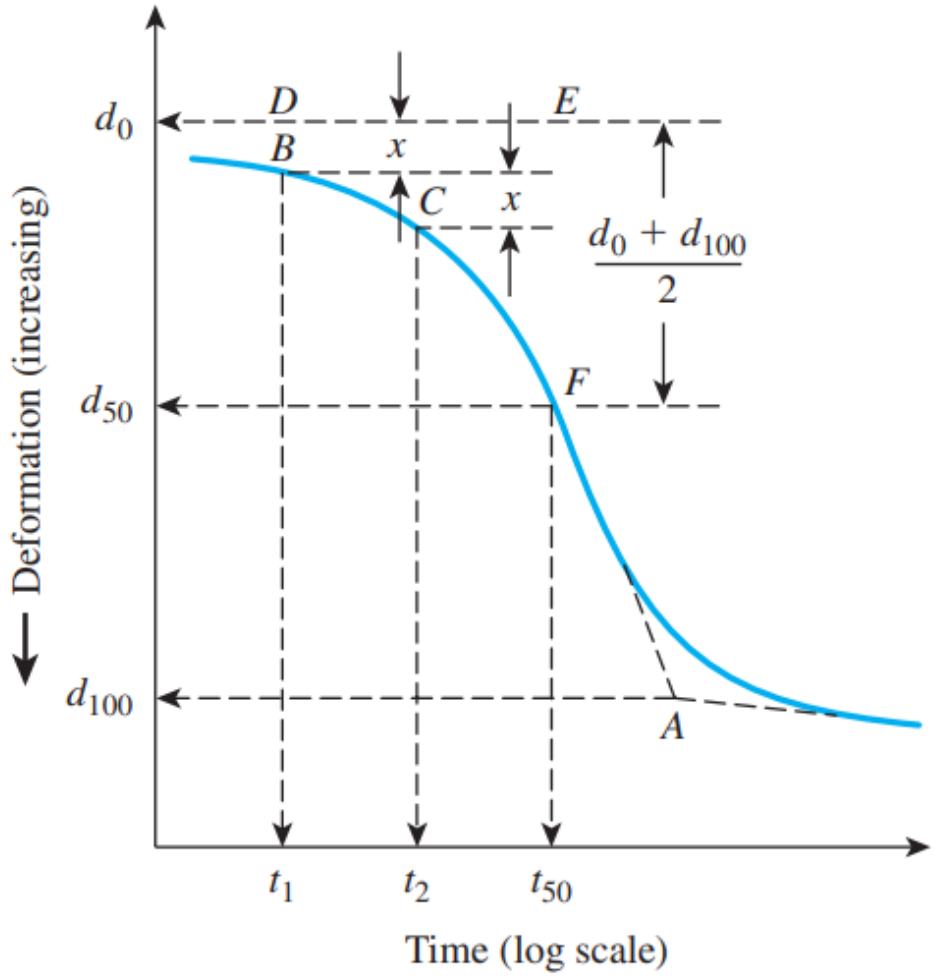
1. Extend the straight-line portions of primary and secondary consolidations to intersect to intersect at A. The ordinate of A is represented by $d_{100}$ - that is, the deformation at the end of 100% primary consolidation.
2. The initial curved portion of the plot of deformation versus log t is approximated to be a parabola on the natural scale. Select time $t_1$ and $t_2$ on the curved portion such that $t_2 = 4t_1$. Let the difference of sample deformation during time $(t_2 - t_1)$ be equal to x.



# Logarithm-of-time method

- Construction procedure

3. Draw a horizontal line DE such that the vertical distance BD is equal to x. The deformation corresponding to the line DE is $d_0$ (that is deformation at 0% consolidation).
4. The ordinate of point F on the consolidation curve represents the deformation at 50% primary consolidation, and its abscissa represents the corresponding time($t_{50}$).



# Logarithm-of-time method

- Construction procedure

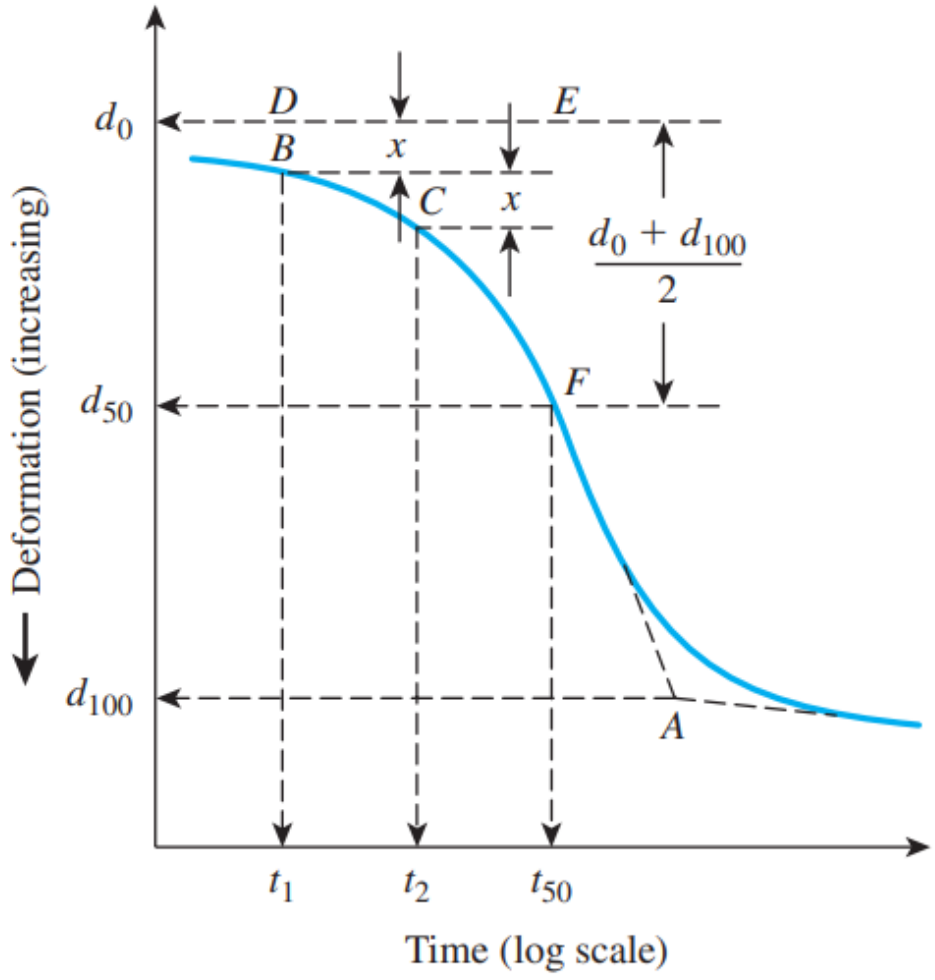
5. For 50% average degree of consolidation,

$$T_v = 0.197 \; or \; 0.196$$

$$T_{50} = \frac{c_v t_{50}}{H_{dr}^2}$$

$$c_v = \frac{0.197 H_{dr}^2}{t_{50}}$$

- For specimens drained at both top and bottom, $H_{dr}$ equals one half of the average height of the specimen.
- For specimens drained on only one side, $H_{dr}$ equals the average height of the specimen during consolidation.



# Square-Root-of-Time Method

- Construction procedure

1. Draw a line AB through the early portion of the curve.
2. Draw a line AC through that $\overline{OC} = 1.15\overline{OB}$. The abscissa of point D, which is the intersection of AC and the consolidation curve, gives the square root of time for 90% consolidation $\sqrt{t_{90}}$
3. For 90% consolidation, $T_{90} = 0.848$, so

$$T_{90} = 0.848 = \frac{c_v t_{90}}{H_{\text{dr}}^2}$$

$$c_v = \frac{0.848 H_{\text{dr}}^2}{t_{90}}$$

